VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING

REPORT
CAPSTONE PROJECT
SEMESTER 242 ACADEMIC YEAR 2024-2025

FINANCIAL DISTRESS PREDICTION FOR ENTERPRISES

MAJOR :COMPUTER SCIENCE
COUNCIL :COMPUTER SCIENCE - CLC 02
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HO CHI MINH CITY, May 2025

HA2 CLC KHMT - 142

ĐẠI HỌC QUỐC GIA TP.HCM
TRƯỜNG ĐẠI HỌC BÁCH KHOA

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

KHOA:KH & KT Máy tính
BỘ MÔN: KHOA HỌC MÁY TÍNH

NHIỆM VỤ LUẬN VĂN/ ĐỒ ÁN TỐT NGHIỆP
Chú ý: Sinh viên phải dán tờ này vào trang nhất của bản thuyết trình

HỌ VÀ TÊN: PHAN THÀNH THÔNG — MSSV: 2153846
NGÀNH: KHOA HỌC MÁY TÍNH — LỚP: CC21KHM2

1. Đầu đề luận văn/ đồ án tốt nghiệp:
Financial distress prediction for enterprises

2. Nhiệm vụ (yêu cầu về nội dung và số liệu ban đầu):
Input: enterprises' financial statements

Output: analyze, predict the financial distress ability of enterprises, and explain its factors

Task:

Analyze financial statement data to classify which firms are going to face financial distress
Forecast the possibility for early warning using machine learning.
Analyze the factors that greatly affect the financial distress of enterprises before and after COVID-19
Present the results on dashboard

3. Ngày giao nhiệm vụ: 06/01/2025

4. Ngày hoàn thành nhiệm vụ: 25/05/2025

5. Họ tên giảng viên hướng dẫn: — **Phần hướng dẫn:**

1) PGS. TS. Quản Thành Thơ — Hướng dẫn thực hiện đồ án, báo cáo

Nội dung và yêu cầu LVTN/ ĐATN đã được thông qua Bộ môn.

Ngày 10 tháng 05 năm 2025
CHỦ NHIỆM BỘ MÔN
(Ký và ghi rõ họ tên)

Trương Tuấn Anh

GIẢNG VIÊN HƯỚNG DẪN CHÍNH
(Ký và ghi rõ họ tên)

Quản Thành Thơ

PHẦN DÀNH CHO KHOA, BỘ MÔN:

Người duyệt (chấm sơ bộ):

Đơn vị:

Ngày bảo vệ:

Điểm tổng kết:

Nơi lưu trữ LVTN/ĐATN:

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TRƯỜNG ĐẠI HỌC BÁCH KHOA
KHOA KH & KT MÁY TÍNH

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

Ngày 10 tháng 05 năm 2025

PHIẾU ĐÁNH GIÁ LUẬN VĂN/ ĐỒ ÁN TỐT NGHIỆP

(Dành cho người hướng dẫn/phản biện)

1. Họ và tên SV: Phan Thành Thông
MSSV: 2153846 Ngành (chuyên ngành): Khoa học máy tính
2. Đề tài: Financial distress prediction for enterprises

3. Họ tên người hướng dẫn/phản biện: PGS.TS. Quản Thành Thơ

4. Tổng quát về bản thuyết minh:
Số trang: 73 Số chương: 5
Số bảng số liệu: 7 Số hình vẽ: 15
Số tài liệu tham khảo: 64 Phần mềm tính toán: Không
Hiện vật (sản phẩm): Không
5. Những ưu điểm chính của LV/ ĐATN:
– Xác định các phương pháp phù hợp nhất để dự đoán khó khăn tài chính giữa các công ty niêm yết trên Sở giao dịch chứng khoán Việt Nam.
– Sử dụng bộ dữ liệu các công ty niêm yết trên Sở giao dịch chứng khoán Việt Nam đầy đủ hơn những nghiên cứu khác ở Việt Nam
– Áp dụng nhiều thuật toán học máy cho nhiều mô hình khác nhau
– Phân tích các chỉ số tài chính để xác định chỉ số nào có tác động lớn nhất đến kết quả dự đoán
– Sử dụng bảng thông tin trong Power BI để trực quan hóa kết quả dự đoán khó khăn tài chính cung cấp một cách rõ ràng và tương tác để phân tích các chỉ số và xu hướng chính

6. Những thiếu sót chính của LV/ĐATN:
– Dự án nên xem xét việc kết hợp các phương pháp dự đoán khó khăn tài chính thay thế
– Chưa tích hợp nhiều thuật toán học máy truyền thống, lai và học sâu hơn để cải thiện hiệu suất dự đoán
– Thiếu dữ liệu về thị trường cũng như nội bộ doanh nghiệp để tiến hành phân tích sâu và chính xác hơn
– Tổng thể giải pháp cho vấn đề chưa có sự khác biệt quá lớn về hiệu quả so với các nghiên cứu khác

7. Đề nghị: Được bảo vệ ☑ Bổ sung thêm để bảo vệ □ Không được bảo vệ □
8. Các câu hỏi SV phải trả lời trước Hội đồng:

9. Đánh giá chung (bằng chữ: Xuất sắc, Giỏi, Khá, TB): Điểm : 8.5/10

Ký tên (ghi rõ họ tên)

Quản Thành Thơ

HA2 CLC KHMT- 142

TRƯỜNG ĐẠI HỌC BÁCH KHOA
KHOA KH & KT MÁY TÍNH

Ngày tháng năm

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

PHIẾU ĐÁNH GIÁ LUẬN VĂN/ ĐỒ ÁN TỐT NGHIỆP

(Dành cho người hướng dẫn/phản biện)

1. Họ và tên SV: Phan ThànhThông MSSV: 2153846 Ngành (chuyên ngành): KHMT

2. Đề tài: Financial distress prediction for enterprises

3. Họ tên người phản biện: Nguyễn Quốc Minh

4. Tổng quát về bản thuyết minh:
Số trang: 61 Số chương: 5
Số bảng số liệu: 7 Số hình vẽ: 15
Số tài liệu tham khảo: 64 Phần mềm tính toán:
Hiện vật (sản phẩm)

5. Những ưu điểm chính của LV/ ĐATN: Sử dụng các mô hình học máy truyền thống (DT, RF, SVM) dựa trên mô hình Altman's Z-score and Zmijewski's X-score để đánh giá khả năng tài chính của các công ty niêm yết trên sàn chứng khoán HOSE/HNX. Có sử dụng những công cụ trực quan hóa.

6. Những thiếu sót chính của LV/ĐATN: LVTN chưa làm rõ vì sao lựa chọn mô hình Altman's Z-score and Zmijewski's X-score, và mô hình này có phù hợp với điều kiện Việt Nam hay không.

7. Đề nghị: Được bảo vệ ☒ Bổ sung thêm để bảo vệ O Không được bảo vệ O

8. Các câu hỏi SV phải trả lời trước Hội đồng:
a. Những khó khăn, thách thức lớn nhất khi thực hiện LVTN?

b. Vì sao lựa chọn Altman's Z-score and Zmijewski's X-score?

c. Mô hình này có còn phù hợp trong bối cảnh hiện tại nữa hay không?

9. Đánh giá chung (bằng chữ: Xuất sắc, Giỏi, Khá, TB): Điểm : 8.6/10

Ký tên (ghi rõ họ tên)

Nguyễn Quốc Minh

Declaration of Authenticity

I confirm that this research is my original work and has been completed independently, carried out under the supervision and guidance of Assoc. Prof. Quan Thanh Tho at the Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology - Vietnam National University.

All sources of information, ideas, and data that are not my own have been properly acknowledged and cited. I understand that plagiarism, or any other form of academic dishonesty, violates institutional policies and ethical principles in academic work.

Furthermore, I certify that this research has not been submitted elsewhere, either in full or in part, to satisfy the requirements of another academic or professional assignment.

By signing and submitting this declaration, I take full responsibility for the originality, integrity, and plagiarism-free nature of this work. Ho Chi Minh City University of Technology - Vietnam National University Ho Chi Minh City bears no responsibility for any copyright infringements that may occur within this research.

Ho Chi Minh City, May 2025

Author,

Phan Thanh Thong

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Abstract

The issue of financial distress in listed companies has become a topic of significant discussion from both academic and practical perspectives, particularly because financial markets continue to play an important role in promoting growth for both local and global economies. In the era of advanced technology and the digital revolution, forecasting and early identification of financial distress are important methods for building investor confidence and enabling prompt decision-making to prevent bankruptcy and extreme financial risk.

This study applies classical machine learning algorithms (Decision tree, Random forest, Support vector machine) to assess the likelihood of financial distress among non-financial firms listed on the Vietnam Stock Exchange, using a dataset comprising 10826 observations from 2010 to 2023. The research identifies internal factors, e.g., ROA%, Debt-to-Equity Ratio, and Total Debt to Total Assets, as key indicators with the most significant impact on financial distress across various financial distress models. The findings show that model Zmijewski X-score (1984) efficiently predicts financial distress with an accuracy rate when using decision tree is 99,63%. However, Random Forest consistently performs the best in both models, showing excellent AUC scores with both all and selected features. It is the most robust model across different configurations. This research adds to the current body of literature.

Table of Glossary

Term	**Definition**	**Note**
Financial distress	a situation where a company is unable to generate enough revenue to meet their financial obligations	FD
Financial distress prediction	predict whether a firm is under financial distress situation	FDP
Artificial Intelligence	a broad field that aims to create machines capable of performing tasks that typically require human intelligence	AI
Machine Learning	a field of AI that focuses on enabling computers to learn from data without being explicitly programmed	ML
SHapley Additive exPlanations	a game theoretic approach to explain the output of any machine learning model	SHAP
Discriminant analysis	a statistical method to find a linear combination of features that separates classes.	DA
Multivariate discriminant analysis	a statistical method that finds linear combinations of variables to best separate groups based on their means and internal variations.	MDA
Supervised learning	a type of machine learning where algorithms learn from labeled data to predict outcomes	
Linear Regression	a ML algorithm models the relationship between variables using a straight line.	LR
Decision Trees	a tree-like model to make decisions based on input features.	DTs

Term	**Definition**	**Note**
Support Vector Machines	a ML algorithm classify data by finding the optimal separating hyperplane	SVMs
Random Forest	a ML algorithm combines multiple decision trees to improve accuracy and reduce overfitting	RF
K-Nearest Neighbors	a ML algorithm classifies data based on the labels of the closest points	KNN
The F1 score	a machine learning evaluation metric that measures a model's accuracy	
Area Under the Receiver Operating Characteristic Curve	a metric used in machine learning to evaluate the performance of binary classification models	AUC-ROC

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Chapter 1

Introduction

Chapter 1 provides an overview of the problem's motivation, outlines the project's objectives and scope, and presents the structure of the report.

1.1 Motivation

For many decades, a substantial body of literature has developed on firm default prediction and corporate financial distress, driven by the critical need to understand and forecast companies' financial conditions, particularly during times of economic turbulence [1]. The world economy and commercial systems were dramatically impacted by the Covid-19 pandemic. It caused waves of company collapses in several sectors and nations, leaving many businesses in financial trouble. This had a major effect on the world economy and led to severe economic distress [2].

Vietnam was not immune to the impact of the Covid-19 pandemic. The prolonged crisis in 2021 exposed the country to significant economic risks. According to reports from Vietnam's General Statistics Office (GSO), the third quarter of 2021 marked the first-ever negative GDP growth in Vietnam since the early 2000s [3]. Although the country ended 2021 with a positive GDP growth rate of 2.58%, this figure was considerably lower than the growth rates of the previous 22 years. The GSO reports also highlighted that the pandemic, along with strict lockdowns and extended social distancing measures, severely impacted many businesses in Vietnam. As many as 119,800 firms left the market in 2021, up 17.8 per cent year-on-year due to the negative impact of Covid-19. Meanwhile, the number of newly established firms dropped to around 160,000, a 10.7% decline from the previous year [4].

Financial distress refers to a situation where a company or individual is unable to generate enough revenue or income to meet their financial obligations or cover expenses [5]. Financial distress occurs when organizations face difficulties in meeting their financial commitments, which frequently results in insolvency or bankruptcy. Early warning systems for financial distress, corporate failure, or bankruptcy represent a important field of research in corporate finance, with financial distress prediction at its core [6]. Financial distress prediction generally uses statistical, mathematical, or intelligent models to analyze current financial data in order to predict whether a company will face financial distress. It has become an increasingly critical area of research and application, especially in the current unstable economy. If financial distress predictions are true, corporate management can take remedial action to prevent further decline before a crisis arises, whereas investors can use the crisis to examine the financial health of listed companies and alter their investment strategy to maximize profit.

Because financial distress is so important to enterprises, the economy, and all relevant stakeholders, it has been a compelling research topic for the past 5 decades [7]. However, most studies on financial distress prediction have been conducted in the contexts of the United States, China, and Europe. This topic remains relatively new in emerging countries, including Vietnam. Vietnam is undergoing deep integration with both the region and the world, presenting a golden opportunity for Vietnamese firms to expand, but also posing significant challenges. As a developing economy, most Vietnamese firms are small and medium-sized, which results in limited investment opportunities, low competitiveness, and an underdeveloped capital market. Very few studies on financial distress prediction using machine learning have been done in the Vietnamese context. For instance, Ha et al. (2023) [8] used machine learning algorithms to calculate the likelihood that listed firms on the Vietnam Stock Exchange would experience financial distress; using the dataset of Vietnamese listed companies, Tran et al., 2022 [9] compared the predictive capabilities of machine learning algorithms and used SHAP values to interpret the prediction results.

1.2 Goals

Approaching the topic of financial distress forecasting through machine learning is essential from both theoretical and practical standpoints. By identifying the early signs of financial distress, firms can respond quickly to minimize costs and address the issue more effectively. This work aims to determine the most suitable methods for predicting financial distress among listed firms on the Vietnam Stock Exchange.

This work applies traditional distress prediction models to identify firms experiencing financial distress then employs machine learning algorithms to predict the likelihood of financial distress. Next, financial indicators will be analyzed to determine which have greatest impact on the prediction outcomes. Following this, managers can propose innovative solutions that yield promising results. Finally, the research summarizes the key findings by visualizing on dashboard, outlines the next steps for further exploration, and suggests potential directions for future research.

1.3 Scope

This study aims to address key questions by analyzing data from 1,665 listed companies on the HOSE, HNX, and UpCom exchanges from 2010 to 2023, using Python and supervised machine learning algorithms to identify financial distress:

- Do traditional distress prediction models, when combined with machine learning algorithms, have the capability to predict financial distress and provide early warning signs of bankruptcy for firms?
- Which traditional distress prediction model, Altman Z-score (1968) or Zmijewski X-score (1984) model, can more accurately predict financial distress in Vietnamese companies?
- Which machine learning algorithm (Decision tree, Random forest, Support vector machine) as well as financial indicators, are the most effective in predicting financial distress in Vietnamese companies?
- What are the differences of the financial distress status before, during, and after the financial crisis Covid-19?

1.4 Thesis Structure

This research consists of five chapters.

Chapter 1 provides an overview of the motivation behind the problem, along with the objectives and scope of the project.

Chapter 2 presents the context of financial statements, financial distress, traditional distress prediction models, machine learning models, key features for financial distress prediction, and the evaluation metrics utilized in this research.

Chapter 3 reviews related works on the same topic, discussing existing methods to provide a deeper understanding of their approaches and limitations.

Chapter 4 outlines the proposed workflow for replicating a baseline model from the selected related work, including a basic demonstration, the resulting outcomes, and visualizations.

Chapter 5 provides a summary of the entire research, highlights its limitations, and outlines the plan for future development.

Chapter 2

Theoretical Background

Chapter 2 provides an overview of financial statements, financial distress, conventional distress prediction models, machine learning approaches, essential features for financial distress prediction, and the evaluation metrics applied in this study.

2.1 Financial Statements

A financial statement is a report that details a company's financial condition and performance. It is created according to publicly established regulations, made available to the public, and can only be altered within strict limits. Issued at regular intervals, financial statements are comparable across companies due to standardized accounting policies and Generally Accepted Accounting Principles (GAAPs) [10].

2.1.1 Balance sheet

The balance sheet is a straightforward report showing a company's assets, liabilities, and its book value or net worth, also referred to as owners' equity or shareholders' equity. It is a listing of all accounts with their balances calculated at a specific point in time, providing a financial snapshot [11].

2.1.2 Financial ratios

Financial ratios are numerical indicators calculated from a company's financial statements, used to assess its performance, financial stability, and operational effectiveness. By comparing different figures from the financial statements, these ratios offer valuable insights into various dimensions of the business [11]. These ratios will serve as independent variables in the financial distress models discussed in **Section 2.3**.

2.2 Financial distress

The term ”financial distress” is intentionally broad and somewhat vague. In general, it refers to the inability to meet obligations (e.g., debt) when they are due. Operational definitions of financial distress typically focus on two key events: bond default and bankruptcy. If a company goes through a period where it is unable to pay its debts, bills, and other obligations by their due dates, it is likely facing financial distress [5].

A company is considered to be in default if it encounters certain issues such as bankruptcy, failure to pay maturing bonds, having overdrawn bank accounts, or being unable to distribute dividends to preferred shareholders [12].

Whereas Altman and Hotchkiss imply when financial distress becomes more severe, a business may enter a state of illiquidity, and in more critical cases, default. This occurs when the company is unable to repay the principal or interest on debts as they become due or when it breaches the terms of its credit agreements [13].

Some indicators that a business is facing default or bankruptcy include filing for bankruptcy, being declared bankrupt either voluntarily or involuntarily under legal regulations, or being delisted from the stock exchange for three consecutive years [14].

The majority of research on financial distress forecasting has concentrated on predicting bankruptcy. However, new research indicates that not all businesses experiencing financial difficulties are going to submit for bankruptcy, demonstrating that financial distress and bankruptcy are not the same thing [15].

2.2.1 Financial distress prediction (FDP)

Early warning systems for financial distress, corporate failure, or bankruptcy are a key area of research in corporate finance, with FDP being at its core. FDP aims to forecast whether a company will experience financial distress based on current financial data, using mathematical, statistical, or intelligent models. It is also referred to as financial failure discrimination, bankruptcy prediction, business failure prediction, and corporate failure prediction, among other terms [6].

According to Pindado and Rodrigues, by ignoring the warning signs of financial distress and the impact it can have on a business’s stability and growth, both domestic and foreign companies have suffered serious repercussions. Numerous businesses have seen significant increases in their financial stability and a successful decrease in their risk of bankruptcy by employing business failure prediction models [16].

2.2.2 Traditional distress prediction models

The two most popular traditional methods for evaluating credit risk in listed firms are accounting-based and market-based models. Market-based models, like the Merton models, and accounting-based models, like the Altman Z-score, Zmijewski X-score and emerging market score models, are frequently used to calculate the default probability for listed companies in a particular nation.

2.2.2.1 Altman Z-score (1968)

The Altman Z-Score model, developed in 1968 by American finance professor Edward Altman, is a numerical tool designed to estimate the likelihood of a company facing bankruptcy within the next two years. It serves as an indicator of a firm's financial stability. Altman employs multivariate discriminant analysis (MDA) to derive linear equations based on financial ratios, which are used to assess whether a firm is likely to face bankruptcy or not.

Related variables:

Market value: is the sum of a firm's outstanding shares.

Book value: also known as shareholders' equity or net asset value, showed in the firm balance sheet, is the value of total assets minus total debts.

Total assets: is the sum of all the assets and debts that a business, individual, or group possesses as of a given date.

Turnover: is the rate at which a business replaces its assets over a given time frame.

Total retained earnings: is the total net profits of a business after dividend payments

Market value: also known as net working capital (NWC), is the difference between a company's current assets, such as cash, accounts receivable (or unpaid bills from customers), and inventories of raw materials and completed goods, and its current liabilities, such as debts and accounts payable.

An alternative to the conventional z-score in statistics, the Altman Z-score is based on five financial ratios that can be computed using information from a company's annual 10-K report. It determines whether a business has a high likelihood of going bankrupt by looking at factors like profitability, leverage, liquidity, solvency, and activity.

The original Z-score [17] designed for public manufacturing companies is shown below:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5 \tag{2.1}$$

Where:

X_1 = Working capital / Total assets

X_2 = Retained earnings / Total assets

X_3 = Profit before tax and interests (EBIT) / Total assets

X_4 = Equity market value / Total debts

X_5 = Revenue / Total assets

The financial distress variable is given a value of 1 if the Z-score is less than 1.81, indicating that the company is in the financial distress zone. If not, it is given a value of 0.

2.2.2.2 X-Score of Zmijewski (1984)

Developed by Zmijewski in 1984, the Zmijewski model is an extension of Altman's model that uses financial analysis to evaluate the debt performance and liquidity of a business. Zmijewski employed probit analysis on a sample of 400 non-bankrupt and 40 bankrupt businesses in his study, developing a model with leverage, return on assets (ROA), and liquidity ratios as important variables.

Zmijewski X-score (1984) [18] formula is shown below:

$$X = -4.3 + 4.5X_1 - 5.7X_2 + 0.004X_3 \tag{2.2}$$

Where:

X_1 = Net profit / Total assets

X_2 = Total debts / Total assets

X_3 = Current assets / Current debts

The X-score above 0 indicates that the company is likely experiencing financial distress, assigning a value of 1 to the financial distress variable. On the other hand, the X-score below 0 suggests that the company is expected to avoid bankruptcy, and thus the financial distress variable is assigned a value of 0.

2.3 Features for FDP

FDP involves using various financial and non-financial features to assess the likelihood of a company facing financial distress, such as bankruptcy or insolvency. The goal is to identify key indicators that can signal potential financial trouble, allowing companies, investors, and creditors to take preventative actions. These features typically come from a company's financial statements (balance sheet, income statement, cash flow statement) and market-based information.

EBIT Margin

Earnings before interest and taxes (EBIT) is a measure of a company's profitability, calculated by subtracting expenses, excluding interest and taxes, from revenue. Also referred to as operating earnings, operating profit, or profit before interest and taxes, EBIT provides insight into a company's core operating performance.

$$\text{EBIT} = \text{Net Income} + \text{Interest} + \text{Taxes}$$

[19]

The operating margin assesses a company's profitability by showing how much profit it generates from each dollar of sales after covering variable production costs like wages and raw materials but before accounting for interest or taxes. It is determined by dividing operating income by net sales. Generally, a higher operating margin indicates greater operational efficiency and the company's ability to convert sales into profit effectively.

$$\text{EBIT Margin} = \frac{\text{Net Income} + \text{Interest} + \text{Taxes}}{\text{Revenue}}$$

[20]

EBITDA Margin

The acronym EBITDA stands for earnings before interest, taxes, depreciation, and amortization. The EBITDA margin is a measure of a company's operating profit as a percentage of its revenue [21].

$$\text{EBITDA Margin} = \frac{\text{Earnings Before Interest and Tax} + \text{Depreciation} + \text{Amortization}}{\text{Revenue}}$$

[22]

2.3.1 Solvency

Current ratio

The current ratio is a liquidity metric used to evaluate a company's capacity to meet short-term obligations due within one year. It provides insight to investors and analysts on how effectively a company can utilize its current assets to cover its current liabilities and other short-term payables.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

[23]

Quick ratio

The quick ratio is a measure of a company's short-term liquidity, evaluating its ability to fulfill immediate obligations using its most liquid assets.

$$\text{Quick Ratio} = \frac{\text{Current Assets} - \text{Inventories}}{\text{Current Liabilities}}$$

[24]

Cash ratio

The cash ratio is a liquidity metric that assesses a company's capacity to meet its short-term obligations solely with cash and cash equivalents.

$$\frac{\text{Cash} + \text{Cash Equivalents}}{\text{Current Liabilities}}$$

[25]

WCTA

Working capital (also known as net working capital or NWC) refers to the difference between a company's current assets and current liabilities. It serves as an indicator of a company's liquidity and short-term financial health, reflecting its capacity to fund day-to-day operations and handle financial challenges or seize opportunities.

$$\text{Working Capital} = \text{Current Assets} - \text{Current Liabilities}$$

[26]

The working capital ratio, as known as Working Capital-to-Total Assets Ratio (WCTA), is a fundamental liquidity metric that gauges a company's ability to meet its short-term financial obligations. It serves as an indicator of the company's overall financial solvency and its capacity to cover current liabilities with its current assets.

$$\text{WCTA} = \frac{\text{Current Assets} - \text{Current Liabilities}}{\text{Total Assets}}$$

[27]

Debt-to-equity ratio

The debt-to-equity (D/E) ratio measures the relationship between a company's total liabilities and its shareholder equity, providing insight into the company's reliance on debt for financing. This ratio helps assess the financial leverage and risk associated with the company's capital structure.

$$\text{D/E ratio} = \frac{\text{Total Liabilities}}{\text{Shareholder Equity}}$$

[28]

2.3.2 Operational Efficiency

Receivables turnover

The accounts receivable turnover ratio is a financial metric that measures how efficiently a company collects outstanding receivables from its customers. It indicates the number of times a company can turn its accounts receivable into cash over a specific period, reflecting its effectiveness in managing credit and collections.

$$\text{Receivables turnover} = \frac{\text{Net Revenue}}{\text{Receivables}}$$

[29]

Inventory turnover

The inventory turnover ratio is a financial metric that assesses how efficiently a company manages its inventory. It is calculated by dividing the cost of goods sold (COGS) by the average inventory value over a specific period. This ratio indicates how often a company sells and replaces its inventory during that time frame, helping to evaluate inventory management efficiency.

$$\text{Inventory turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

[30]

Accounts payable turnover

The accounts payable turnover ratio is a financial metric that measures the speed at which a company settles its short-term liabilities. It calculates how frequently a company pays off its accounts payable during a specific period, providing insights into the efficiency of its payment practices and short-term liquidity.

$$\text{AP Turnover} = \frac{\text{TSP} \times (\text{BAP} + \text{EAP})}{2}$$

Where:

AP = Accounts payable

TSP = Total supply purchases

BAP = Beginning accounts payable

EAP = Ending accounts payable

[31]

Asset turnover

Asset turnover is a financial ratio that compares a company's total sales or revenue to its average assets. This metric helps investors assess how efficiently a company utilizes its assets to generate revenue, providing insight into the company's operational effectiveness.

$$\text{Asset turnover} = \frac{\text{Net Sales}}{\text{Average total assets}}$$

[32]

Current assets turnover

The current asset turnover ratio is similar to the asset turnover ratio but is specifically calculated by dividing a company's net sales by its average current assets, rather than total assets. This ratio measures how efficiently a company uses its current assets, such as cash, inventory, and receivables, to generate sales.

$$\text{Current assets turnover} = \frac{\text{Net Sales}}{\text{Current assets}}$$

2.3.3 Profitability

Return on Assets (ROA)

Return on Assets (ROA) is a financial metric that measures a company's profitability in relation to its total assets. It helps corporate management, analysts, and investors assess how effectively a company utilizes its assets to generate profits. A higher ROA indicates more efficient use of assets in generating earnings.

$$\text{ROA} = \frac{\text{Net Income}}{\text{Total assets}}$$

[33]

Return on Equity (ROE)

Return on Equity (ROE) is a financial metric that evaluates a company's financial performance by dividing net income by shareholders' equity. Since shareholders' equity represents a company's assets minus its liabilities, ROE provides insight into how effectively a company generates profits from its net assets. A higher ROE indicates better utilization of equity to generate earnings.

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' equity}}$$

[34]

Return on Invested Capital (ROIC)

Return on Invested Capital (ROIC) is a financial metric used to evaluate a company's effectiveness in using capital to generate profitable returns. It is calculated by dividing Net Operating Profit After Tax (NOPAT) by invested capital. ROIC helps investors assess how well a company is utilizing its capital to produce profits and is an important indicator of overall business efficiency.

$$\text{ROIC} = \frac{\text{Net operating profit after tax}}{\text{Invested capital}}$$

[35]

Gross Profit Margin

Gross profit margin is a financial metric that is calculated by subtracting the cost of goods sold (COGS) from a company's net sales. It is often expressed as a percentage of net sales, representing the proportion of sales revenue that exceeds the cost of goods sold. This margin indicates how much profit a company makes before accounting for other expenses like selling, general, and administrative costs, which are included when calculating the net profit margin.

$$\text{Gross Profit Margin} = \frac{\text{Gross profit}}{\text{Net Revenue}}$$

[36]

Operating Profit Margin

The operating margin indicates the profit a company earns from each dollar of sales after covering variable production costs, such as wages and raw materials, but before accounting for interest or taxes. It is calculated by dividing operating income by net sales. A higher operating margin is generally viewed as positive, as it demonstrates the company's efficiency in its operations and its ability to convert sales into profits.

$$\text{Operating Profit Margin} = \frac{\text{Operating Profit (EBIT)}}{\text{Net Revenue}}$$

[20]

Net Profit Margin

Net profit margin, also known as net margin, is a financial metric that measures the percentage of revenue that remains as profit after all expenses, taxes, and costs have been deducted. It is calculated by dividing net profit by total revenue. This ratio reflects how efficiently a company is able to convert revenue into actual profit, providing insight into its overall profitability.

$$\text{Net Profit Margin} = \frac{\text{Net Profit}}{\text{Net Revenue}}$$

[37]

2.3.4 Growth Capability

Capital Expenditure Growth YoY

Capital expenditures (CapEx) refer to the funds a company uses to acquire, upgrade, or maintain physical assets like property, plants, buildings, technology, or equipment. These expenditures are typically made to support new projects or investments. Examples of CapEx include repairing a roof to extend its useful life, purchasing new equipment, or constructing a new factory.

$$\text{CapEx} = \Delta\text{PP\&E} + \text{Current Depreciation}$$

Where:

CapEx = Capital expenditures

ΔPP&E = Change in property, plant, and equipment

[38]

Revenue Growth 3Y

Three-yearly revenue growth refers to the increase in a company's sales over a three-year period compared to the sales of the previous three-year period. It is an important metric used to evaluate a company's long-term performance and financial health, providing insights into sustained growth and the effectiveness of its business strategies [39].

Profit Before Tax (PBT) Growth 3Y

Profit Before Tax (PBT) is a financial metric that represents a company's profitability before accounting for corporate income tax expenses. It reflects the total profit generated by the company from its operations, excluding tax considerations, and is used to assess the company's financial performance [40].

Net Profit Growth 3Y

Net profit growth refers to the increase in a company's net profit over a specific period. Net profit is the amount remaining after all expenses, including operating costs, interest, taxes, and depreciation, have been deducted from total revenue. It is a key indicator of a company's overall financial health and profitability [41].

2.3.5 Risk Level

Total Debt on Total Assets

The debt ratio, also known as the debt-to-assets ratio, is a financial metric that measures the proportion of a company's total debt relative to its total assets. It is calculated by dividing the company's total debt by its total assets. This ratio helps assess a company's leverage and indicates how much of the company's assets are financed through debt. A higher debt ratio suggests a higher level of leverage, which could mean more financial risk.

$$\text{Total Debt on Total Assets} = \frac{\text{Short-Term Debt} + \text{Long-Term Debt}}{\text{Total Assets}}$$

[42]

2.3.6 Others

Basic Earnings Per Share (EPS)

Earnings per share (EPS) is a financial metric that measures a company's profitability, showing how much profit is allocated to each outstanding share of common stock. It is calculated by dividing the company's net income by the total number of outstanding shares. EPS is widely used by investors to assess a company's financial performance and profitability on a per-share basis, helping to evaluate the company's ability to generate profit relative to its stock.

$$\text{EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{End-of-Period Common Shares Outstanding}}$$

Where:

- EPS = Earnings per share
- Net Income = The net income of the company
- Preferred Dividends = Dividends paid to preferred shareholders
- End-of-Period Common Shares Outstanding = The number of common shares outstanding at the end of the period

[43]

Basic Price-to-Earnings (P/E)

The price-to-earnings (P/E) ratio is a financial metric that compares a company's share price to its earnings per share (EPS). Often referred to as the price or earnings multiple, the P/E ratio is used to evaluate the relative value of a company's stock. It is particularly useful for comparing a company's current valuation to its historical performance, its peers within the same industry, or the broader market. A higher P/E ratio may suggest that the stock is overvalued or that investors expect high growth in the future, while a lower P/E ratio may indicate undervaluation or slower growth prospects.

$$\text{Basic P/E} = \frac{\text{Market value per share}}{\text{Earnings per share}}$$

[44]

Price-to-Book (P/B)

The price-to-book (P/B) ratio is a financial metric that investors use to compare a company's market capitalization to its book value, helping to identify potentially undervalued companies. The ratio is calculated by dividing the company's current stock price per share by its book value per share (BVPS).

$$\text{P/B} = \frac{\text{Market price per share}}{\text{Book value per share}}$$

[45]

Enterprise Multiple (EV/EBITDA)

The enterprise multiple, also known as the EV multiple, is a financial ratio used to assess the value of a company. It is calculated by dividing the enterprise value (EV) by earnings before interest, taxes, depreciation, and amortization (EBITDA). This ratio is important because it evaluates a company from the perspective of a potential acquirer, taking into account both its market value and debt. The enterprise multiple helps compare companies of different sizes and capital structures, offering insight into how much an investor is paying for each dollar of EBITDA. What is considered a ”good” or ”bad” enterprise multiple can vary depending on the industry, with different sectors having different norms for valuation.

$$\text{Enterprise Multiple} = \frac{\text{EV}}{\text{EBITDA}}$$

Where:

EV = Enterprise Value = Market capitalization + Total debt - Cash and cash equivalents

EBITDA = Earnings Before Interest, Taxes, Depreciation, and Amortization

[46]

Dividend yield

The dividend yield is a financial ratio that indicates the annual dividend payment a company makes to its shareholders relative to its stock price. It is calculated by dividing the annual dividend per share by the stock’s current market price. This ratio helps investors assess the return on their investment through dividends. The reciprocal of the dividend yield is known as the dividend payout ratio, which measures the proportion of a company’s net income that is paid out as dividends to shareholders.

$$\text{Dividend Yield} = \frac{\text{Annual Dividends Per Share}}{\text{Price Per Share}}$$

[47]

2.4 Machine learning algorithms

Machine learning (ML) is a subset of computer science and artificial intelligence (AI) that focuses on developing methods that let computers learn and improve over time. The goal is to replicate human learning processes, which will lead to ongoing gains in task accuracy. Machine learning's main objectives are using well-established models to classify data (e.g., identifying spam emails) and using these models to forecast future events (e.g., predicting real estate prices in a city) [48].

Language translation, consumer preference prediction, and medical diagnosis including financial distress prediction are just a few of the many uses for machine learning.

Supervised learning

One of the fundamental techniques in artificial intelligence and machine learning is supervised machine learning. It involves employing labeled data to train a model, where each input is matched with the appropriate output. This process is known as ”supervised” learning because it resembles a teacher giving advice to a student [49].

Types of supervised learning:

Classification refers to the task of predicting a categorical label, such as determining whether a firm is in financial distress or not, or diagnosing a medical condition based on a set of symptoms. Some commonly used algorithms for classification tasks include Logistic Regression, Decision Trees, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Neural Networks, etc.

Regression involves predicting a continuous numerical value. For instance, estimating the price of a house based on factors such as its size, location, and the number of rooms. Some commonly used algorithms for regression tasks include Linear Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), Neural Networks, etc.

Process of supervised learning:

1. Data Collection: compile the input-output pairs that make up the labeled dataset.

2. Model Selection: depending on the kind of problem (regression or classification), select the best algorithm or model.

3. Training: involves providing the input data to the model and optimizing its parameters by minimizing the errors identified through the loss function. This process aims to improve the model's performance over time.

4. Evaluation: make sure the model generalizes well to new, unseen data, test it on a different test set after training.

5. Prediction: make predictions about fresh input data using the trained model.

2.4.1 Support Vector Machine

Support Vector Machines (SVM) are supervised learning algorithms used for classification and regression tasks. SVM aims to find the best decision boundary (also called the hyperplane) that separates data points from different classes while maximizing the margin between the closest data points (support vectors).

In machine learning, a hyperplane serves as the decision boundary that separates different classes within the feature space; in two dimensions, it appears as a line, while in three dimensions, it is a plane. The margin refers to the distance between the hyperplane and the nearest data points, known as support vectors, which play a critical role in defining the decision boundary. SVM aim to maximize this margin to improve classification accuracy. To handle cases where data is not linearly separable, the kernel trick is employed, projecting the data into higher dimensions using functions such as polynomial or radial basis function (RBF) kernels.

For linear SVM, the equation of a hyperplane in n-dimensional space is:

$$w \cdot x + b = 0$$

Where

w is the normal vector to the hyperplane.

x is the input feature vector.

b is the bias term.

Maximize the margin while ensuring all points are classified correctly:

$$\text{Minimize: } \frac{1}{2}||w||^2$$

Subject to:

$$y_i(w \cdot x_i + b) \geq 1 \quad \forall i$$

[50]

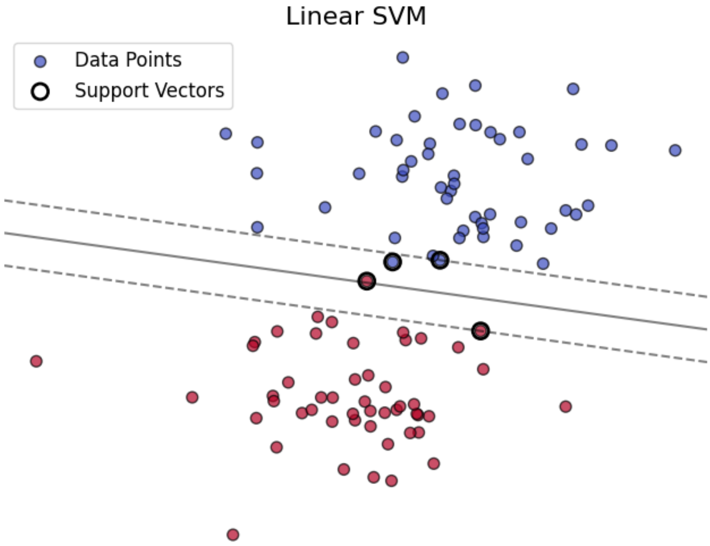

Figure 2.1: *Linear SVM [Author's Source]*

SVM has several strengths, such as their ability to perform well in high-dimensional spaces, making them ideal for situations where the number of dimensions surpasses the number of data points. They are also resistant to overfitting, especially in high-dimensional datasets, due to the margin maximization approach. SVMs are adaptable, using kernel functions to manage non-linearly separable data, and are efficient in memory usage by relying solely on support vectors to define the decision boundary. On the other hand, SVMs can be computationally intensive for large datasets, require careful selection of kernels and parameters for optimal results, and can be difficult to interpret, particularly when non-linear kernels are employed.

2.4.2 Decision Tree

Decision trees (DT), developed by Morgan and Sonquist in 1963 during their search for the determinants of social conditions [51], are widely used and highly effective tools in fields like machine learning, data mining, and statistics. They offer a clear and intuitive method for making decisions based on data by modeling the relationships between various variables. It is a supervised learning algorithm applied to both classification and regression problems. It operates by recursively splitting the data into subsets based on the values of input features, forming a tree-like structure of decisions. The tree is made up of nodes, where decisions are made, and branches, which represent the outcomes of those decisions. Each internal node corresponds to a feature, each branch represents a decision based on that feature, and the leaf nodes indicate the final output labels or values.

DT operate on the following principles:

1. Decision trees use **Recursive Splitting**, where the data is divided at each node based on a specific feature and its value to optimize a chosen criterion, such as Gini impurity or entropy for classification tasks, or variance reduction for regression tasks.

2. The main objective when selecting splits in a decision tree is to choose the one that achieves the maximum reduction in impurity at a given node, typically evaluated using specific metrics:

   Gini Impurity (for Classification)

   $$Gini(D) = 1 - \sum_{i=1}^{C} p_i^2$$

   Where:

   p_i is the proportion of class i in dataset D.

   C is the number of classes.

   Entropy (for Classification)

   $$Entropy(D) = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

   where:

   p_i is the proportion of class i in dataset D.

   $\log_2(p_i)$ is the logarithm of p_i to base 2.

   Variance Reduction (for Regression)

   $$MSE(D) = \frac{1}{|D|} \sum_{i=1}^{|D|} (y_i - \bar{y})^2$$

   where:

   - y_i is the target value of instance i.
   - $\bar{y}$ is the mean target value of dataset D.
   - $|D|$ is the number of data points in D.

   Information Gain (for Classification)

   $$IG(D,A) = Entropy(D) - \sum_{v \in Values(A)} \frac{|D_v|}{|D|} Entropy(D_v)$$

Where:

D is the dataset.

A is the attribute.

$Values(A)$ is the set of all possible values for attribute A.

D_v is the subset of D where the value of attribute A is v.

3. Stopping Criteria:

Maximum depth of the tree.

Minimum samples per leaf or node.

Minimum impurity decrease.

Maximum number of leaf nodes.

[52]

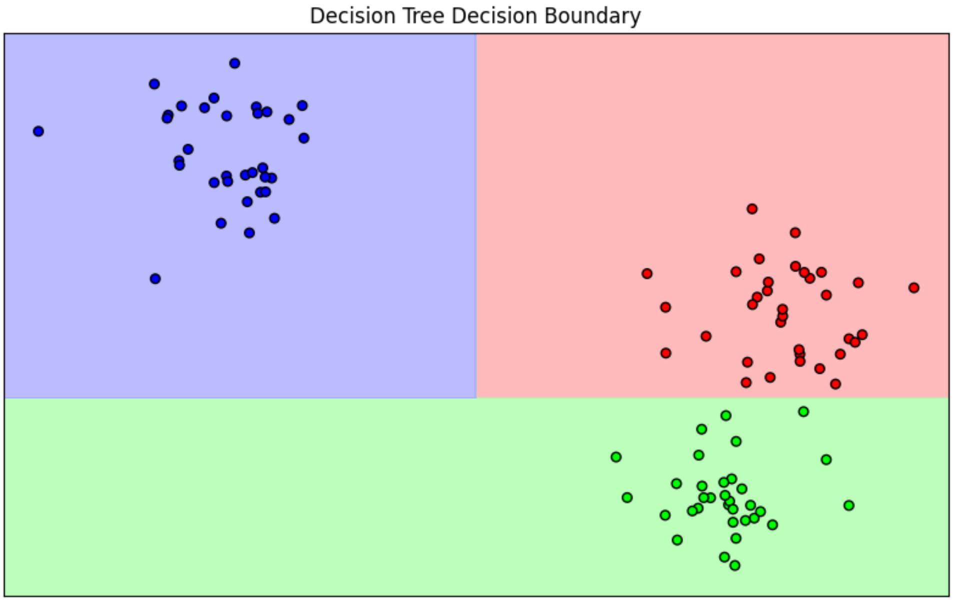

Figure 2.2: *Decision tree decision boundary [Author's Source]*

Decision trees are widely used for classification and regression due to their numerous benefits. Their straightforward and visual structure makes them easy to interpret and understand, simplifying the decision-making process. They are capable of handling non-linear relationships, require no feature scaling, and can effectively process both numerical and categorical data. Moreover, decision trees can assess feature importance, highlighting the most influential variables in predictions. On the other hand, decision trees have some drawbacks. They are susceptible to overfitting, particularly when the tree becomes too complex, and can be sensitive to small data changes, leading to significant alterations in the tree's structure. Additionally, they may demonstrate bias toward features with more levels and rely on a greedy approach, which optimizes each split individually but may not yield the best overall results.

2.4.3 Random Forest

Random Forest (RF) is an advanced ensemble learning technique commonly employed for classification and regression tasks. It operates by constructing multiple decision trees during the training phase and merging their outputs to enhance accuracy and mitigate overfitting. For classification tasks, the final prediction is determined through majority voting among the trees, while for regression, the output is the average of the predictions from all the trees.

RF operates based on several key concepts. One is bootstrap aggregation (bagging), where multiple subsets of the original dataset are created using random sampling with replacement, and each subset is used to train an individual decision tree. Another is random feature selection, where at each split in a decision tree, a random subset of features is evaluated instead of all features, reducing the correlation between trees. Finally, ensemble prediction combines the outputs of all decision trees: for classification tasks, the final class is determined by majority voting, while for regression tasks, the final prediction is the average of all tree predictions.

Mathematical Equations:

Prediction for Classification

$$\hat{y} = \text{mode}\big(T_1(x), T_2(x), \ldots, T_m(x)\big)$$

Where $T_i(x)$ is the prediction of the i-th tree, and m is the total number of trees.

Prediction for Regression

$$\hat{y} = \frac{1}{m}\sum_{i=1}^{m} T_i(x)$$

Gini Impurity (used in splitting trees)

$$Gini(D) = 1 - \sum_{i=1}^{C} p_i^2$$

Entropy (optional for splits)

$$Entropy(D) = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

[53]

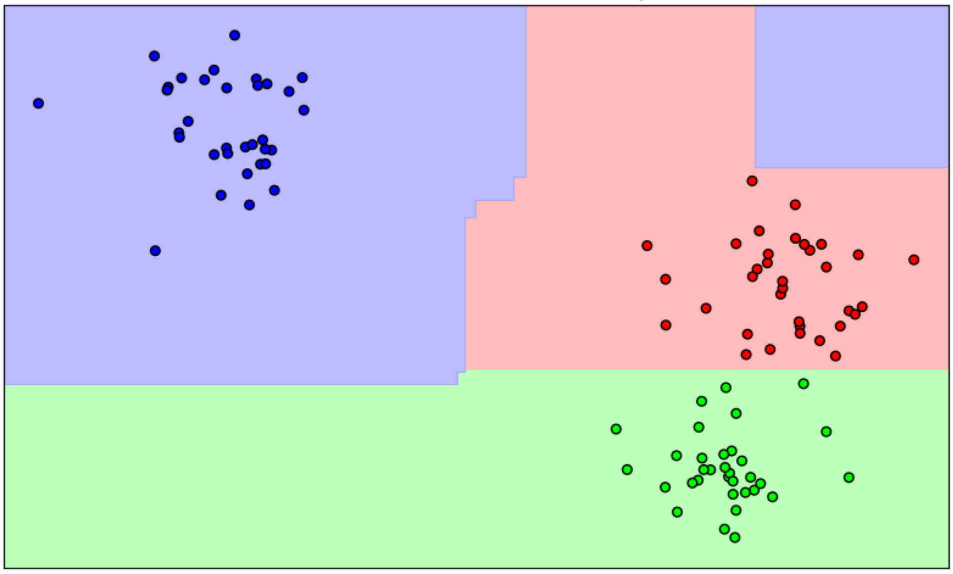

Figure 2.3: *Random forest decision boundary [Author's Source]*

RF offers several advantages that make it a powerful tool for both classification and regression tasks. It tends to provide higher accuracy than a single decision tree by reducing variance through averaging predictions. Its robustness makes it less susceptible to overfitting, as the ensemble method helps smooth out individual errors. Additionally, RF is highly scalable, performing well with large datasets and high-dimensional data. Its versatility allows it to be effectively used for a wide range of problems in both classification and regression domains.

2.4.4 Feature Importance

Mean Decrease in Impurity (MDI) and Permutation Importance are two common methods to evaluate feature importance in tree-based models (e.g., Decision Trees, Random Forests, Gradient Boosting)

2.4.4.1 Mean Decrease in Impurity (MDI) - Gini Importance

MDI uses a measure of impurity, such as Gini impurity or entropy, to quantify how mixed up the target variable is in a node. A node with high impurity has a mix of different classes, while a pure node has only one class. When a decision tree splits a node based on a feature, it aims to reduce the impurity of the resulting child nodes. A good split will create child nodes that are more homogeneous than the original parent node. For each feature, MDI calculates the total reduction in impurity across all trees in the random forest where that feature is used for splitting. This is then averaged across all trees to get the final MDI value. Features with higher

MDI values are considered more important because they contribute more to reducing impurity and improving the model's predictive accuracy [54].

2.4.4.2 Permutation Importance

Permutation Importance works by randomly shuffling the values of a feature and observing how much the model's performance decreases. A feature is considered important if its permutation causes a significant drop in model performance [55].

The model is trained on the original dataset and its performance (e.g., accuracy, AUC) is measured on a hold-out or validation set. This establishes a baseline score. The values of a specific feature are randomly shuffled. This breaks the relationship between that feature and the target variable. The model is evaluated on the permuted data. The difference between the baseline performance and the performance after shuffling is calculated. This difference is the permutation importance score. A larger difference indicates a more important feature. This process is repeated for each feature in the dataset. The features are then ranked based on their permutation importance scores. Features with higher scores are considered more important [56].

2.4.4.3 Comparison

Table 2.1: MDI and Permutation Importance Comparison

Criterion	MDI (Gini Importance)	Permutation Importance
Computation Time	Fast (built-in)	Slow (needs multiple evals)
Model-agnostic	Only for tree models	Yes
Bias toward feature type	Can be biased	Less biased
Handles correlated features	Not well	Not well
Captures interactions	Limited	Yes
Interpretability	Relative importance	Easy to explain

This work uses Permutation Importance because it is more objective, not affected by the number of feature values and easy to explain.

2.5 Evaluation metrics

In machine learning and statistical modeling, evaluation metrics are essential tools for assessing the performance of a model, especially when dealing with classification tasks. These metrics provide insight into how well the model performs in predicting outcomes, with a focus on distinguishing between different classes (e.g., positive/negative, distressed/not distressed). Below is an overview of five widely used evaluation metrics: accuracy, precision, recall, F1 score, and AUC-ROC.

2.5.1 Accuracy

Accuracy is one of the most basic and commonly used evaluation metrics, representing the proportion of correctly predicted instances to the total number of instances.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Instances}}$$

Accuracy is a straightforward metric that provides a quick snapshot of overall model performance. However, it can be misleading, particularly in imbalanced datasets where one class significantly outnumbers the other. In such cases, a model that consistently predicts the majority class may still achieve high accuracy, but this does not necessarily mean it has effectively learned to distinguish between the classes. Therefore, accuracy alone may not always reflect a model's true performance, especially when dealing with class imbalances. Accuracy is best used when the classes are balanced, i.e., the number of instances of each class is roughly equal.

2.5.2 Precision

Precision measures the proportion of positive predictions that were actually correct. It is particularly useful when the cost of false positives is high. For instance, in financial distress prediction, falsely predicting a financially healthy company as distressed (false positive) might result in unnecessary interventions, which could be costly.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Precision focuses on the correctness of positive predictions, measuring how often the model correctly identifies positive cases. A high precision indicates that when the model predicts the positive class (e.g., financial distress), it is likely to be accurate. Precision is especially important in scenarios where the consequences of a false positive are costly or undesirable, as it ensures that only the most relevant cases are flagged as positive, reducing unnecessary errors.

2.5.3 Recall

Recall measures the proportion of actual positives that were correctly identified by the model. It is useful in situations where false negatives (missing an instance of the positive class) are more costly or risky than false positives. In financial distress prediction, failing to identify a distressed company (false negative) could have severe consequences.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Recall emphasizes identifying as many positive instances as possible, aiming to capture all relevant cases. A high recall indicates that most of the actual positive cases (e.g., distressed companies) are detected, but this could lead to including some false positives. Recall is particularly valuable in situations where missing a positive instance, such as failing to identify a distressed company, is unacceptable. In these cases, ensuring that as many true positives as possible are identified takes precedence, even if it results in some incorrect predictions.

2.5.4 F1 Score

The F1 score is the harmonic mean of precision and recall, providing a balance between the two. It is particularly useful when you need a metric that accounts for both false positives and false negatives, especially in imbalanced datasets.

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1 score balances the trade-off between precision and recall, providing a more comprehensive measure when both false positives and false negatives are important. A high F1 score indicates that the model performs well in both identifying positive cases and minimizing false positives. This metric is particularly useful when you need a single evaluation metric that is not overly influenced by class imbalance, offering a more balanced view of the model's performance.

2.5.5 AUC-ROC

The AUC-ROC, Area Under the Receiver Operating Characteristic Curve, is a graphical representation that shows the trade-off between true positive rate (recall) and false positive rate (FPR) across various threshold values. The ROC curve plots the true positive rate against the false positive rate at different classification thresholds, and the AUC measures the area under the curve.

Formula for False Positive Rate (FPR):

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$

AUC-ROC offers a comprehensive measure of model performance across all possible thresholds. A higher AUC value, closer to 1, indicates that the model is better at distinguishing between the positive and negative classes, while a value of 0.5 suggests that the model has no discriminatory power, essentially performing at the level of random guessing. ROC and AUC are valuable for understanding how well the model can separate the classes at different decision thresholds, making them especially useful in situations with imbalanced classes, where the distribution of one class significantly outweighs the other.

2.6 Discussion

The theoretical background of FDP using supervised machine learning classification combines traditional financial distress models with advanced computational techniques to provide more accurate and dynamic predictions. Utilizing machine learning algorithms like decision trees, support vector machines, and random forests allows financial distress prediction to incorporate a broader range of features, resulting in more robust and scalable models compared to traditional statistical methods. Machine learning models frequently leverage a diverse set of input features, and evaluation metrics are employed to assess the performance of each model within FDP models.

Chapter 3

Related Works

Chapter 3 presents a review of previous studies on the same topic, examining current methods to offer a clearer insight into their strategies and constraints.

Financial distress prediction for businesses is fundamentally a binary classification problem, where predictive models are used to categorize companies into normal or at-risk categories. New techniques are continuously being discovered and developed to improve the accuracy of predicting corporate financial distress.

3.1 Traditional financial distress prediction techniques

Discriminant analysis (DA) was the commonly employed method for financial distress prediction prior to 1980, as seen in studies by Altman (1968) and Beaver (1966).

Beaver (1966) was the first to introduce a univariate statistical model that examined the ability of 29 financial ratios to predict corporate financial distress as much as five years in advance [12]. In 1968, Altman developed a multivariate Z-score model, choosing five independent variables to create the Z-score index [17]. Later, in 1980, Ohlson introduced conditional probability regression models to estimate the likelihood of corporate bankruptcy, overcoming the limitation of Z-scores, which lacked economic significance [57].

The Z-score (Altman, 1968) [17] and the X-score (Zmijewski, 1984) [18] are widely used metrics to assess whether a company is in financial distress, as evidenced by numerous recent studies.

3.2 Machine learning approach

Hung & Chen (2009) suggested a selective ensemble consisting of three classifiers: the decision tree, backpropagation neural network, and support vector machine. By leveraging the expected probabilities of both bankruptcy and non-bankruptcy, this ensemble approach combines the strengths of different classification techniques while mitigating their weaknesses[58].

To analyze the impact of Covid-19 on firms' financial distress, Ding et al. (2023) propose a financial distress prediction model based on the Extreme Gradient Boosting-Genetic Programming (XGB-GP) framework, examining subsamples from the pre-COVID and post-COVID periods. Their findings reveal that during the pre-COVID period, the earnings financial indicator is the dominant feature in predicting financial distress, whereas in the post-COVID period, total financial leverage becomes the key factor influencing distress prediction [59].

Cheraghali & Molnár (2024) used the Light Gradient Boosting Machine (LightGBM) and Extreme Gradient Boosting (XGBoost), both of which offer superior predictive accuracy. Moreover, proper validation and feature selection are crucial for enhancing model performance [60]. Wang & Liang (2024) developed the adaptive weighted XGBoost-Bagging model for predicting corporate financial distress. Their empirical findings show that systemic risk indicators have predictive power independent of traditional financial data, making them valuable non-financial early warning signals for China's industrial sector [61].

In the context of Vietnam, several notable studies have applied machine learning techniques to forecast the financial distress of listed firms.

Tran et al. (2022) compared the predictive power of various machine learning algorithms and applied SHAP (Shapley Additive Explanations) values to interpret the prediction results on a dataset of listed companies in Vietnam. The study found that the Extreme Gradient Boosting and Random Forest models outperformed other models in terms of predictive accuracy [9].

Ha et al. (2023) combined traditional financial distress prediction models, such as the Altman Z-Score and Zmijewski X-Score, with machine learning algorithms like Decision Tree and Random Forest. This hybrid approach achieved an impressive accuracy rate of 98% in predicting financial distress, demonstrating the effectiveness of blending traditional models with advanced machine learning techniques for enhanced prediction accuracy [8].

These studies leverage various algorithms such as decision trees, support vector machines, and ensemble methods to predict corporate financial health, focusing on the use of financial ratios and other indicators specific to the Vietnamese market. By utilizing these advanced methods, researchers aim to enhance prediction accuracy and provide valuable early warnings of potential financial distress for firms in Vietnam.

3.3 Discussion

Machine learning has significantly improved the prediction of financial distress by addressing the limitations of traditional models. Studies have demonstrated the superiority of ML techniques such as random forests, SVM, artificial neural networks, and deep learning in handling complex, high-dimensional datasets and capturing non-linear relationships. Despite these advances, challenges such as data quality, model interpretability, and dealing with imbalanced data remain, and future research will likely focus on overcoming these obstacles and developing more robust models for financial distress prediction.

Chapter 4

Proposed Solutions

Chapter 4 outlines the proposed workflow for reproducing a baseline model from the chosen related study, featuring a simple demonstration, the obtained results, and accompanying visualizations.

4.1 Data analytics

Baseline Production

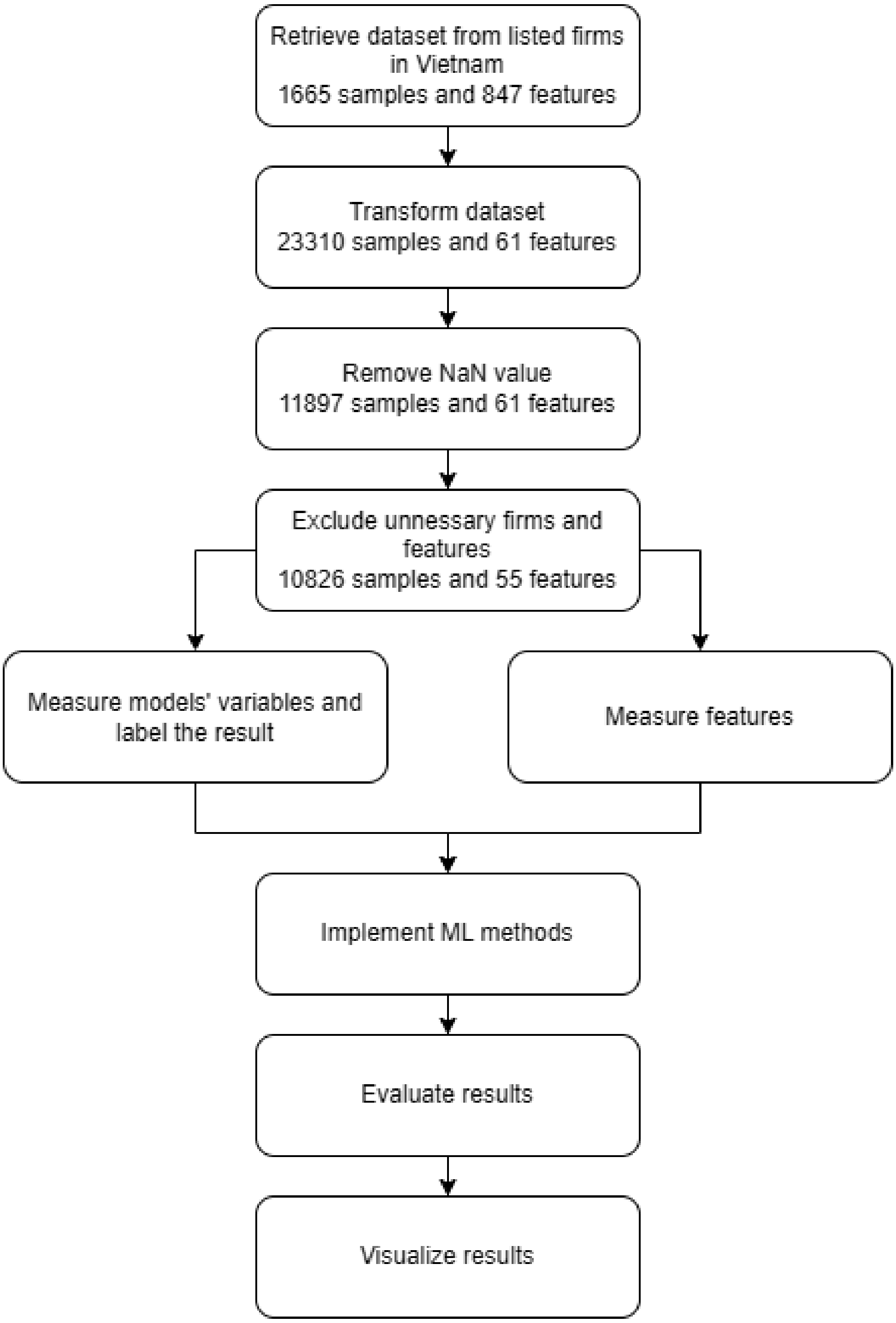

Figure 4.1: *Financial distress prediction workflow [Author's Source]*

4.1.1 Data Retrieval and Pre-processing

This study utilizes data gathered from the Vietnam Stock Exchange spanning the period from 2010 to 2023. The dataset is derived from audited financial statements of listed companies, excluding those in the banking, securities, and insurance sectors due to the distinct characteristics of these industries compared to others. Null values are removed from the dataset because we cannot generate information of a feature based the others.

4.1.2 Variables Measurement

Based on the formulas of Altman Z-score [2.1] and Zmijewski X-score [2.2], evaluate the score of each firms, then label them with 1 (financial distress) and 0 (normal)

Based on the Features for FDP on **Chapter 2** and informantion from the dataset, calculate the value and using them as model input features. The summary of the Features can also be found at **Appendix B**.

4.1.3 Descriptive Statistics

The data spanning from 2010 to 2023 provides a comprehensive look at the distribution of observations over time. A central measure such as the mean percentage across all years can help identify the typical observation percentage, while the median can offer insight into the year with the most representative percentage. The range, from 2010 to 2023, captures a variety of economic conditions, allowing for an analysis of how external factors, like the COVID-19 pandemic, might have influenced the distribution. For example, the data from 2021 shows surprising stability despite global economic challenges, possibly due to factors like increased digitalization and government interventions. This can be explored further by looking at the standard deviation to assess how much fluctuations in yearly percentages deviate from the average.

The dataset's distribution by sector reveals interesting insights into which industries are more represented and how they contribute to the overall observations. The Industrial sector is the largest contributor, accounting for 42.25% of the dataset. This high percentage suggests that industrial firms may either be more prone to financial distress or that the sector is overrepresented in the dataset due to its size and economic importance. On the other hand, the Consumer Goods sector contributes 18.30%, signaling its crucial role in financial distress analysis, as consumer demand and sensitivity to economic fluctuations can significantly impact this sector. Other sectors such as Materials (13.15%), Utilities (9.67%), and Consumer Services (7.98%) also make substantial contributions, indicating their relevance in the context of financial analysis.

Model 1 (Altman's Z-score) shows a balanced classification, with many observations flagged as distressed, suggesting higher sensitivity to distress cases. However, this could lead to false positives. Model 2 (Zmijewski's X-score) appears more conservative, potentially reducing false positives but possibly missing true distress cases. Comparing the "Normal" vs. "Financial Distress" classifications of both models will help evaluate their sensitivity and specificity.

4.1.4 Machine Learning Implementation

By comparing the three machine learning models (Decision Tree, Random Forest, and SVM), I aim to determine which model provides the most efficient and accurate prediction of financial distress. The Decision Tree model is likely to offer a simple, interpretable approach, while the Random Forest model, as an ensemble method, could enhance accuracy by mitigating overfitting. SVM, known for its robustness in high-dimensional spaces, might offer strong performance but could be computationally expensive.

In comparing the machine learning models, we will also consider the trade-offs between false positives and false negatives. Model selection will depend on the specific business goals—whether minimizing false positives (avoiding unnecessary distress flags) or minimizing false negatives (identifying all distressed cases) is the priority. The final choice of algorithm will be based on performance metrics and its alignment with real-world outcomes, ensuring that the selected model is both accurate and practical for predicting financial distress.

4.1.5 Models Evaluation

When evaluating machine learning models for predicting financial distress, it is essential to consider multiple performance metrics to ensure the models' effectiveness and suitability for the task. The three models under evaluation—Decision Tree, Random Forest, and Support Vector Machine (SVM)—each have strengths and weaknesses that may impact their performance.

Metrics like precision, recall, and F1-score will help assess the models' reliability and effectiveness in predicting financial distress. Model 1 may be better for early warnings, while Model 2's cautious approach could prevent unnecessary alarms.

4.1.6 Features Importance

Once the optimal model is selected for further analysis, feature importance using Permutation Importance is assessed to evaluate the impact of each feature on the model. Features contributing less than 0.01 (1%) to the total importance may be considered negligible. Subsequently, the feature set is reduced, and the model's performance is re-evaluated.

4.2 Experimental Result

4.2.1 Evaluation

This figure presents the results of two models, Altman's Z-score (Model 1) and Zmijewski's X-score (Model 2), in predicting financial distress. The percentages reflect the proportion of observations classified as either "Normal" or "Financial Distress" by each model.

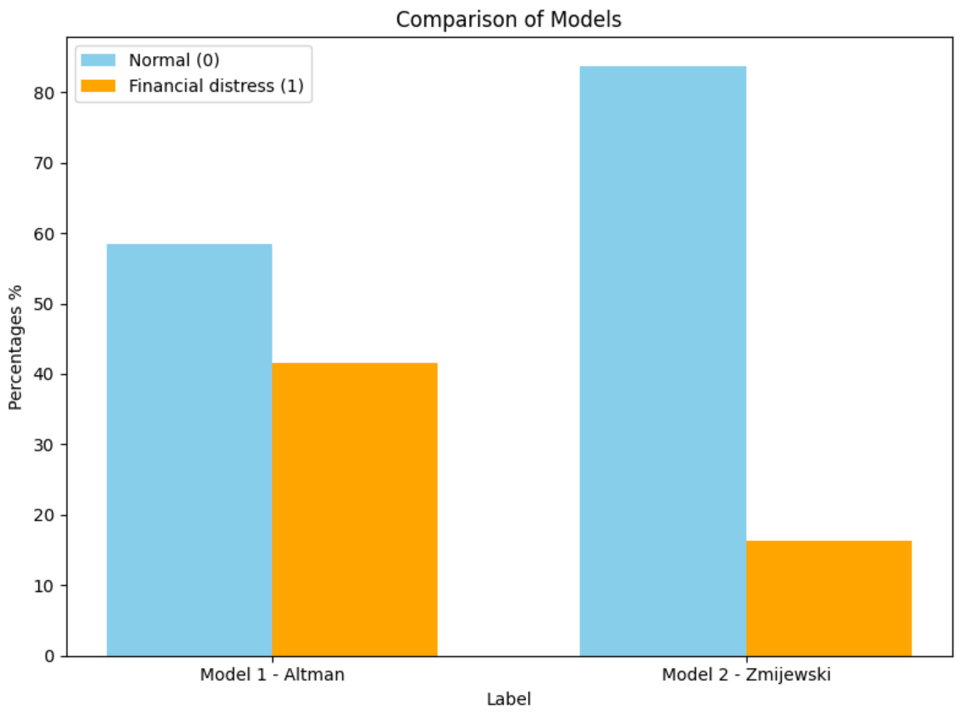

Figure 4.2: *Data by label [Author's Source]*

Model 1 indicates a relatively balanced classification, with a significant portion flagged as distressed, suggesting the model might be more sensitive to financial distress cases. Whereas, Zmijewski model might be more conservative in identifying distress, potentially resulting in fewer false positives but possibly missing some actual distress cases. The distribution of "Normal" and "Financial Distress" cases in the dataset could influence model outputs. It would be helpful to verify if either model aligns more closely with real-world outcomes.

Result of ML algorithms on each financial distress models with **all features** as input:

Table 4.1: ML algorithms evaluation metrics with all features in Model 1 - Altman

ML Model	Accuracy	Precision		Recall		F1	
		0	1	0	1	0	1
Random Forest	0.9353	0.9270	0.9482	0.9652	0.8936	0.9457	0.9201
Decision Tree	0.9135	0.9177	0.9073	0.9356	0.8825	0.9266	0.8947
SVM	0.6268	0.6116	0.8730	0.9873	0.1220	0.7553	0.2140

Table 4.2: ML algorithms evaluation metrics with all features in Model 2 - Zmijewski

ML Model	Accuracy	Precision		Recall		F1	
		0	1	0	1	0	1
Random Forest	0.9926	0.9956	0.9779	0.9956	0.9779	0.9956	0.9779
Decision Tree	0.9951	0.9967	0.9871	0.9974	0.9835	0.9970	0.9853
SVM	0.8319	0.8326	0.2500	0.9989	0.0018	0.9082	0.0036

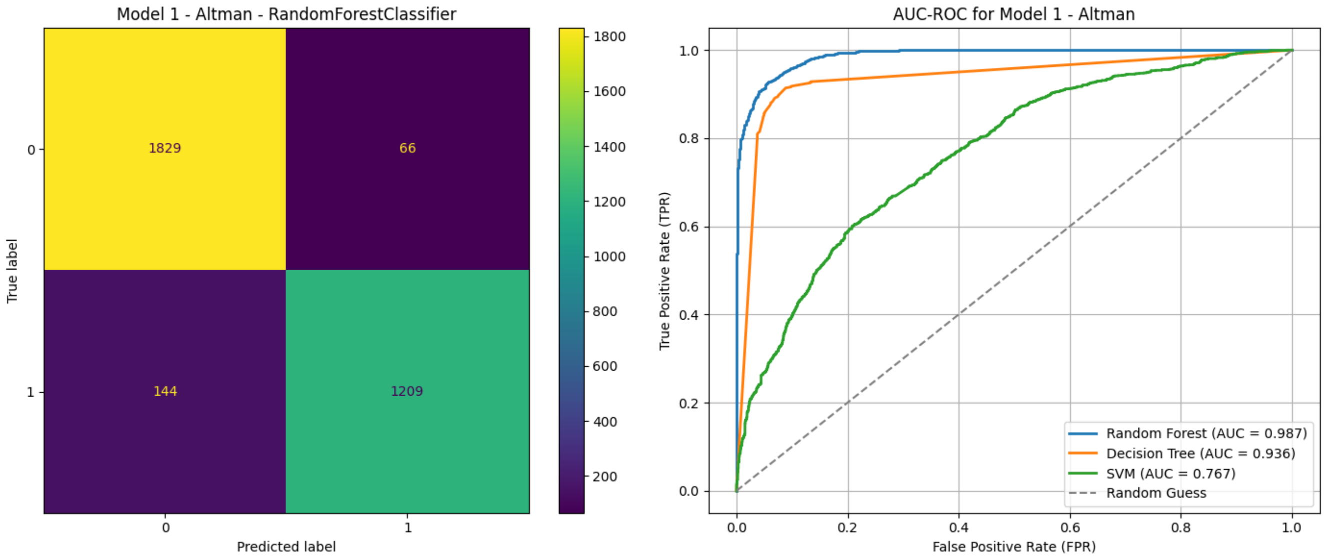

Figure 4.3: *AUC-ROC of Model 1 - Altman with all features according to ML algorithms [Author's Source]*

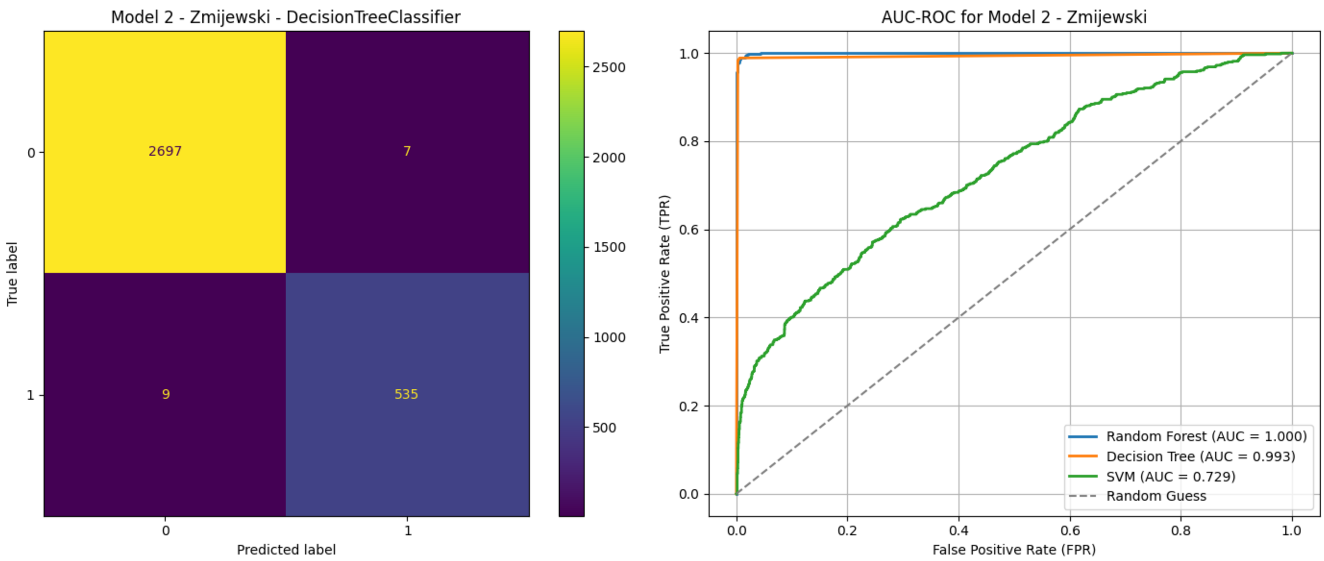

Figure 4.4: *AUC-ROC of Model 2 - Zmijewski with all features according to ML algorithms [Author's Source]*

Random Forest and Decision Tree have provided the best overall performance, particularly for Model 2 - Zmijewski. These models achieve high accuracy, precision, recall, and F1 scores, making them reliable for predicting financial distress.

SVM consistently struggles with imbalanced predictions, especially for Model 1 - Altman and Model 2 - Zmijewski, resulting in poor F1 scores and significant bias towards the positive class. This suggests that SVM might not be suitable for this task without proper tuning or balancing techniques.

Precision and Recall metrics highlight that both Random Forest and Decision Tree effectively handle the detection of financially distressed companies while minimizing the number of false positives and false negatives, making them more suitable for financial distress prediction.

Use RF Permutation Importance to identify the most influential features in the model, any feature contributing less than 0.01 (1%) to the overall importance may be excluded:

Table 4.3: Selected features using RF Permutation Importance in Model 1 - Altman

Feature	**Importance**
X11: Asset turnover	0.174007
X23: Total Debt on Total Assets	0.066975
X26: P/B	0.054804
X12: Current assets turnover	0.035139
X13: ROA %	0.033510
X6: WCTA	0.033245
X17: Operating Profit Margin	0.022887
X7: D/E ratio	0.017933
X3: Current ratio	0.015485
X25: Basic P/E	0.008707

Table 4.4: Selected features using RF Permutation Importance in Model 2 - Zmijewski

Feature	**Importance**
X23: Total Debt on Total Assets	0.220797
X13: ROA %	0.047841
X7: D/E ratio	0.008880

Result of ML algorithms on each financial distress models with **selected features**:

Table 4.5: ML algorithms evaluation metrics with selected features in Model 1 - Altman

ML Model	**Accuracy**	**Precision**		**Recall**		**F1**	
		0	**1**	**0**	**1**	**0**	**1**
Random Forest	0.9367	0.9324	0.9432	0.9612	0.9024	0.9466	0.9223
Decision Tree	0.9159	0.9161	0.9157	0.9422	0.8791	0.9290	0.8970
SVM	0.5844	0.5841	1.0	1.0	0.0022	0.7374	0.0044
XGB	0.9364	0.9363	0.9465	0.9628	0.9035	0.9445	0.9249
ANN	0.9067	0.9261	0.8804	0.9129	0.8980	0.9195	0.8891
CNN	0.8822	0.9145	0.8408	0.8805	0.8847	0.8972	0.8622

Table 4.6: ML algorithms evaluation metrics with selected features in Model 2 - Zmijewski

ML Model	Accuracy	Precision		Recall		F1	
		0	1	0	1	0	1
Random Forest	0.9953	0.9966	0.9887	0.9977	0.9832	0.9972	0.9860
Decision Tree	0.9963	0.9966	0.9943	0.9988	0.9832	0.9977	0.9887
SVM	0.9205	0.9139	0.9895	0.9988	0.5264	0.9545	0.6872
XGB	0.9958	0.9955	0.9971	0.9994	0.9777	0.9975	0.9873
ANN	0.9939	0.9972	0.9779	0.9955	0.9860	0.9963	0.9819
CNN	0.9926	0.9972	0.9698	0.9939	0.9860	0.9955	0.9779

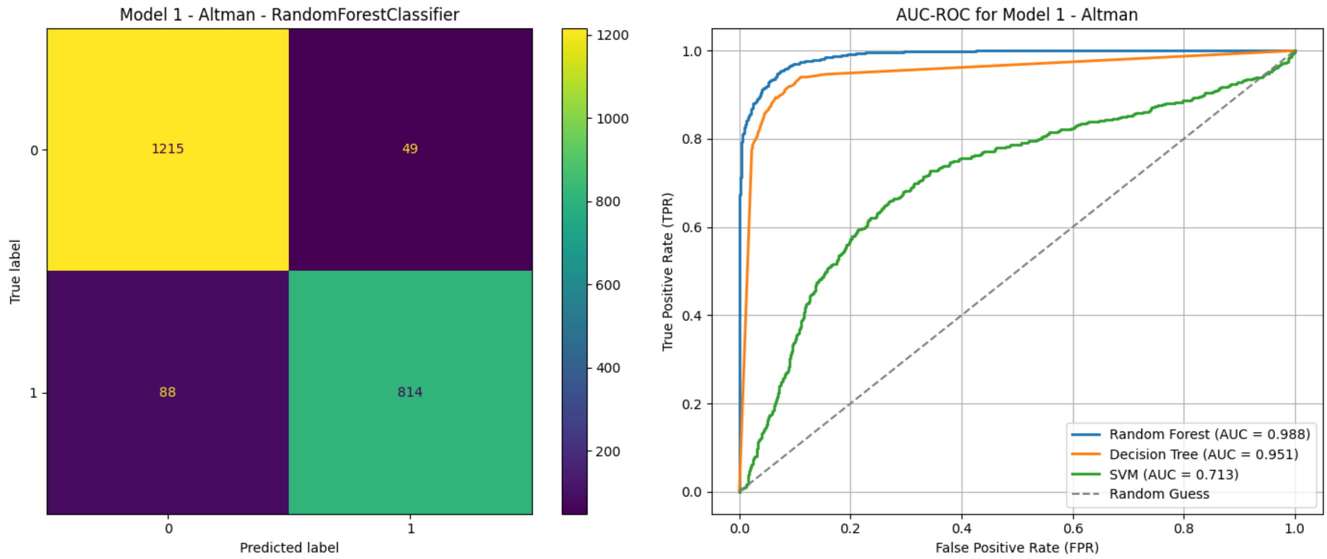

Figure 4.5: *AUC-ROC of Model 1 - Altman with selected features according to ML algorithms [Author's Source]*

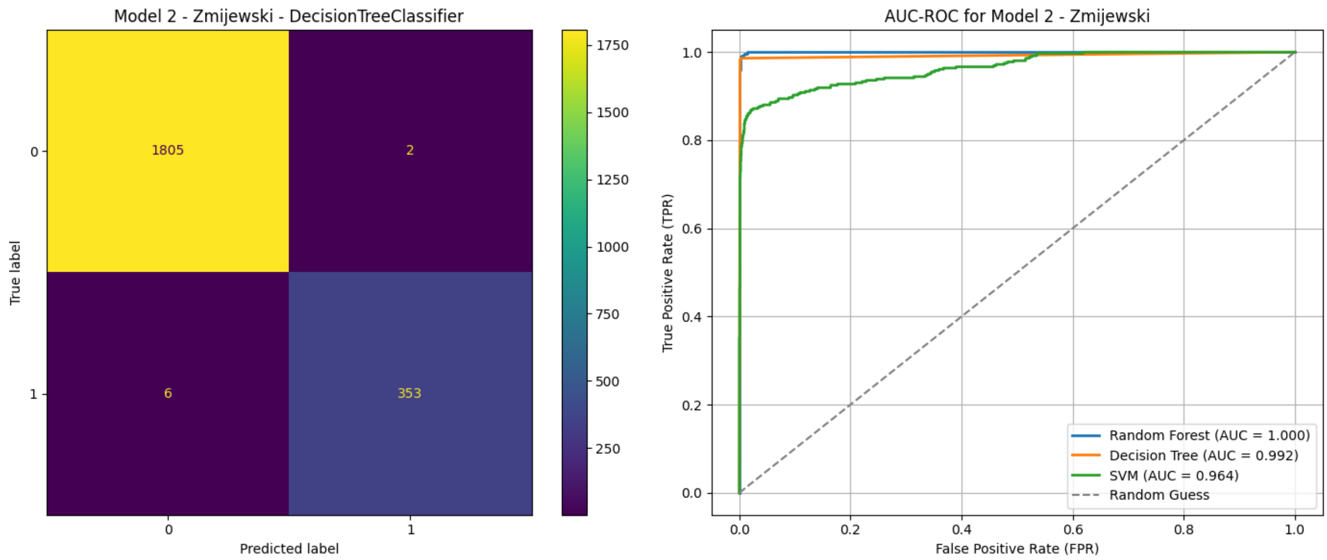

Figure 4.6: *AUC-ROC of Model 2 - Zmijewski with selected features according to ML algorithms [Author's Source]*

Random Forest and Decision Tree perform exceptionally well in both the Altman and Zmijewski models, showing high accuracy, precision, recall, and F1 scores. These models demonstrate strong ability to identify distressed companies while maintaining a good balance between false positives and false negatives.

SVM consistently performs poorly, especially in the Altman model, where its Accuracy drops dramatically. The low F1 (1) scores and the significant imbalance in precision and recall indicate that SVM struggles to make accurate predictions in financial distress prediction.

Selected features have improved the overall performance for the most part, particularly for Random Forest and Decision Tree, where both models show enhanced all scores compared to previous results, suggesting that feature selection has been beneficial.

4.2.2 Visualization

Using a dashboard in Power BI to visualize the results of financial distress prediction provides a clear and interactive way to analyze key indicators and trends. These visuals help users quickly identify companies at risk of financial distress. By analyzing historical data and prediction results, users can observe patterns over time, evaluate the effectiveness of predictive models, and make informed strategic decisions. Overall, the dashboard transforms complex data into actionable insights for financial analysts and decision-makers.

The dashboards below are based on the results from Model 1 – Altman with the Random Forest Classifier. This model was chosen because it demonstrates relatively balanced classification performance and includes many features with significant importance values.

4.2.2.1 Dashboard - Overview

The ”Overview” dashboard presents a basic analysis of the dataset and the financial distress status over the years. It includes information on sectors, trading floors, debt-to-asset and profit comparisons, as well as the number of financial distress firms by year, categorized by market capitalization.

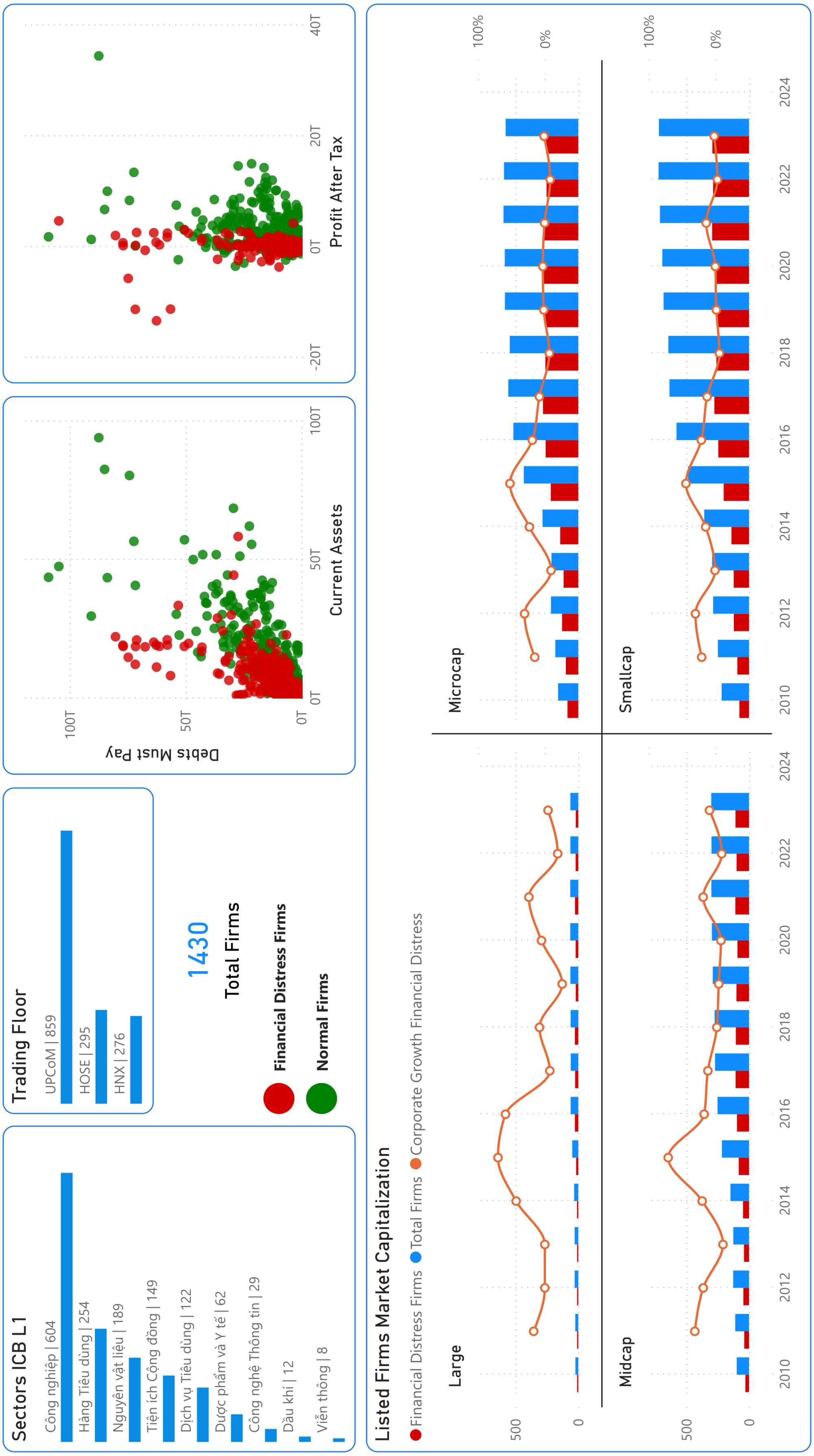

Figure 4.7: *FDP Dashboard - Overview*

Let's examine each component of the dashboard in detail:

This figure represents the distribution of observations by industry sectors (ICB L1 classification) and trading floors:

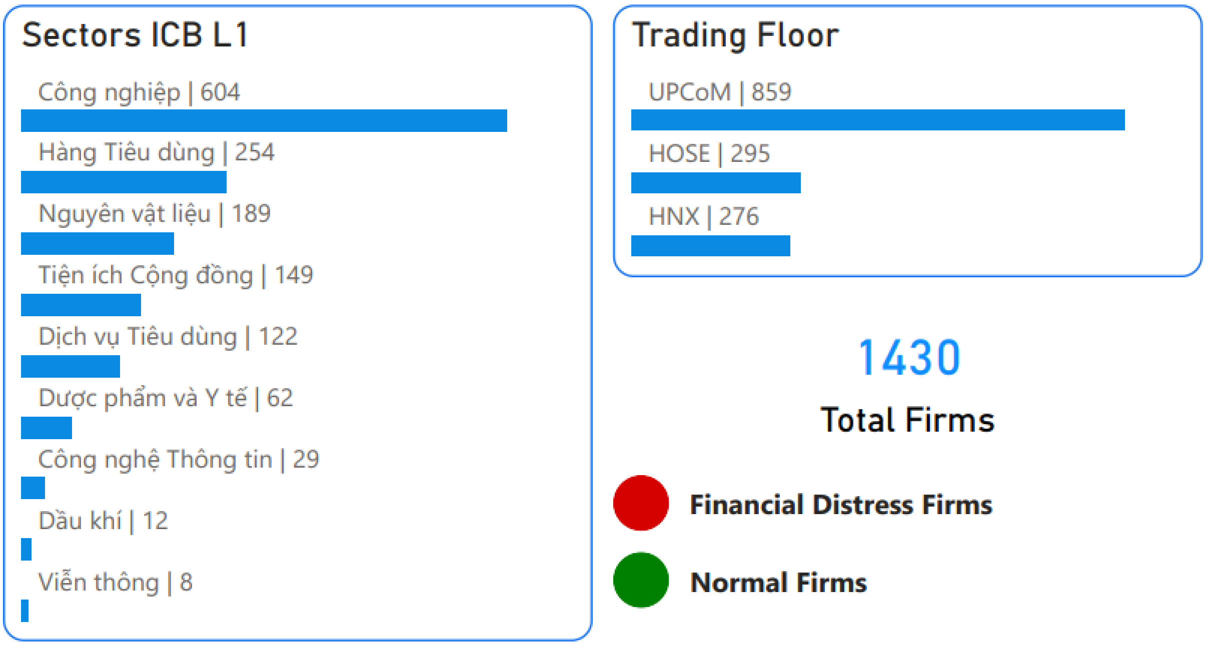

Figure 4.8: *Sector & Trading Floor*

The Industrial sector contributes 604 observations, making it the largest segment in the dataset. This prominence may indicate a higher vulnerability to financial distress within the sector or reflect its dominance in the dataset due to scale or relevance.

The Consumer Goods sector follows with 254 observations, underscoring its importance in financial distress analysis. Its inclusion may be attributed to the sector's dependence on consumer demand and its responsiveness to economic changes.

Other notable contributors include the Materials sector (189 observations), Utilities (149), and Consumer Services (122). These sectors likely appear prominently due to their significant economic roles and potential exposure to financial instability.

The UPCoM market dominates in terms of the number of listed firms. This is often due to it being a less stringent listing venue compared to HOSE and HNX, allowing smaller or newer companies to participate. In contrast, HOSE and HNX are more selective, typically listing larger or more established companies. This distribution suggests that a significant portion of firms in Vietnam's stock market are still in early development stages or have not yet met the requirements for HOSE or HNX.

This figure represents a scatter plot comparing debt-to-asset ratios with profit levels, providing a visual representation of the relationship between a firm's financial leverage and its profitability:

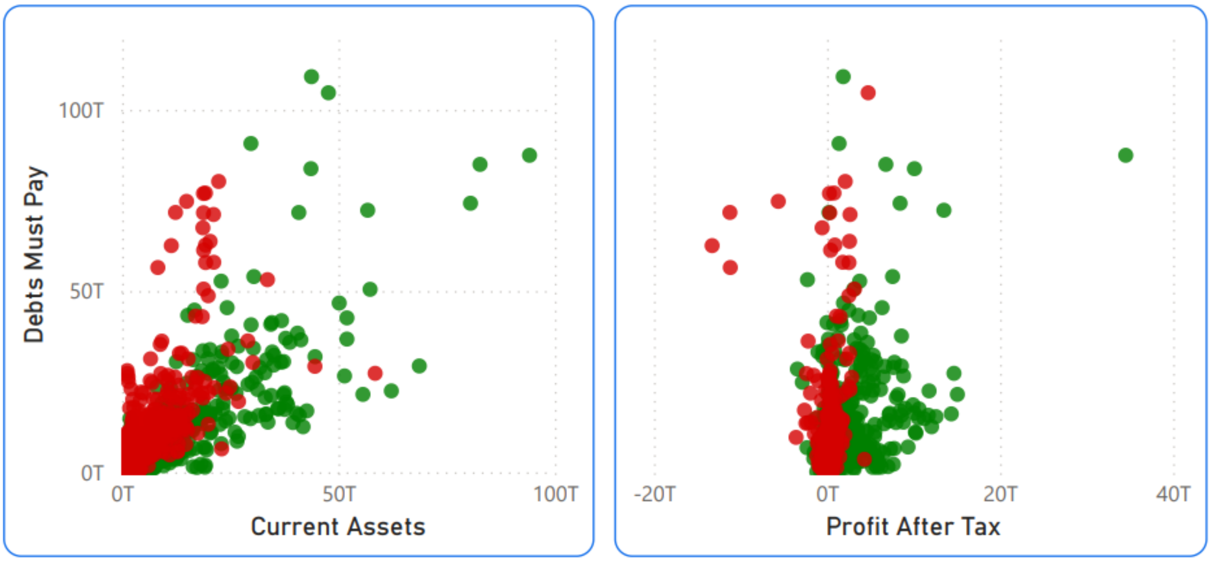

Figure 4.9: *Debt-to-asset and profit comparisons*

The charts highlight that financially distressed firms generally suffer from low liquidity (low current assets vs. high debt) and poor profitability (negative or low profit after tax). In contrast, normal firms maintain healthier financial ratios, with better asset-to-debt balance and positive earnings.

This figure shows the annual count of financially distressed firms, broken down by market capitalization categories, offering insights into how financial distress trends vary across different company sizes over time:

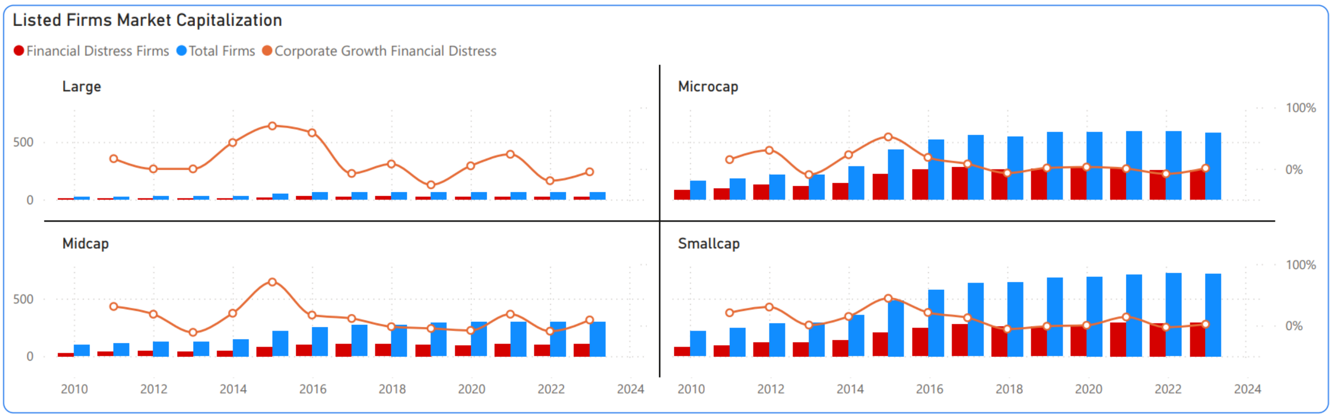

Figure 4.10: *The number of financial distress firms by year categorized by market capitalization*

The "Listed Firms Market Capitalization" chart highlights that large-cap firms exhibit the greatest resilience—with relatively few in distress and volatile but recoverable growth—while mid-caps occupy a middle ground, showing both vulnerability and occasional rebounds. In contrast, micro- and small-cap firms face persistent financial distress: their distressed counts have risen sharply, and their growth rates remain flat or negative. Overall, the data reveal a strong link between firm size and financial health, suggesting that smaller firms may need more targeted support to withstand economic shocks.

The data for 2021 shows a relatively stable percentage of observations compared to previous years, despite the global crisis of COVID-19. This stability might seem surprising, but several potential explanations could account for this: increased digitalization during the pandemic, economic policies and financial monitoring, etc.

4.2.2.2 Dashboard - Performance Ratio

Each chart in the dashboard displays a performance ratio over the years, based on the table of selected features shown in **Table 4.3**.

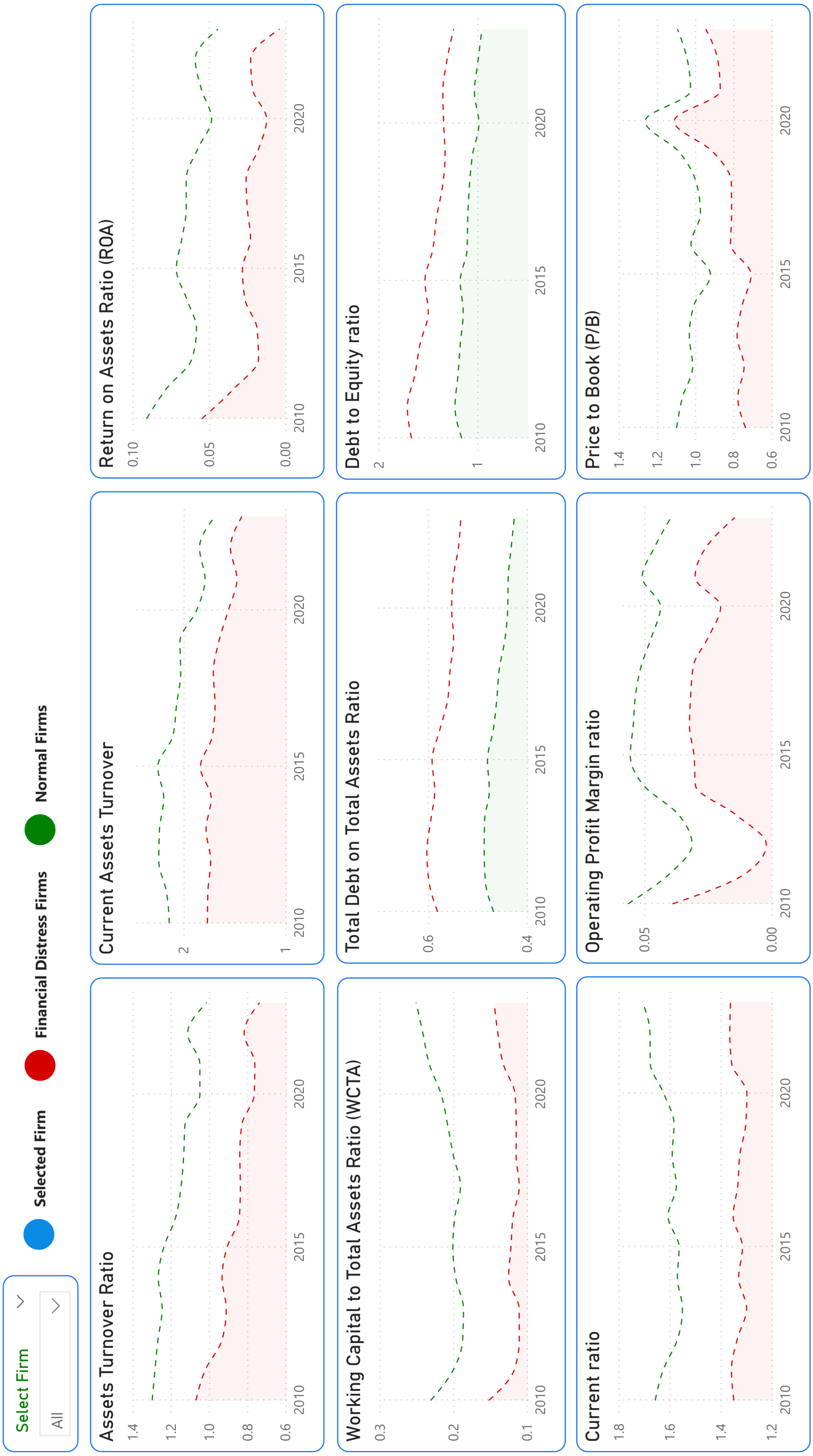

Figure 4.11: *FDP Dashboard - Performance Ratio*

The dashboard can also display information for each firm in the dataset. For instance:

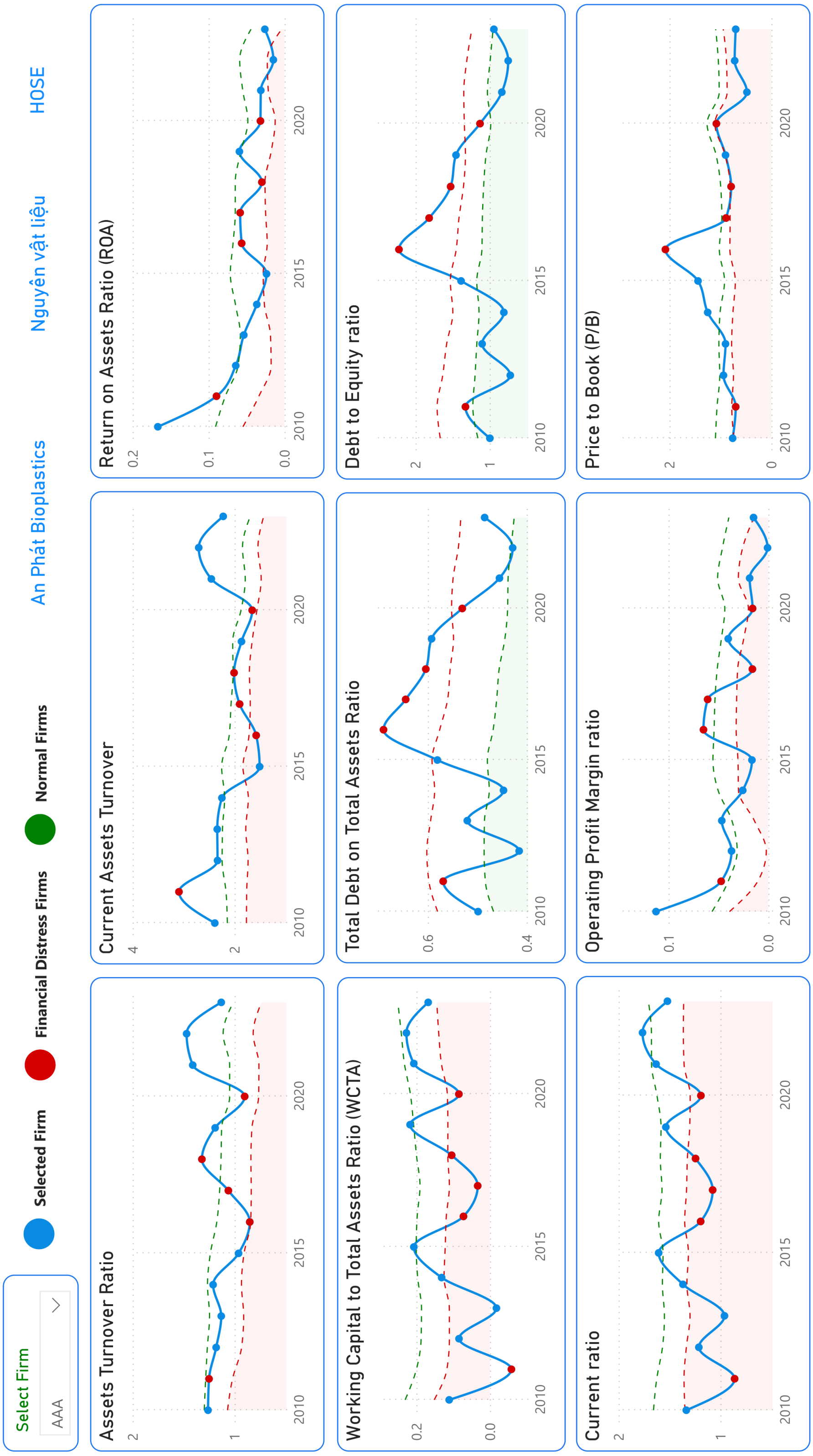

Figure 4.12: *FDP Dashboard - Performance Ratio with Selected Firm*

The red point represents the financial distress status of the selected firm of that year.

4.3 Dicussion

Model 2 (Zmijewski) is more effective than Model 1 (Altman) for predicting financial distress, particularly in the Random Forest and Decision Tree models. It demonstrates better AUC scores, higher accuracy, and greater consistency in performance across different feature sets. Therefore, Model 2 is the superior choice for predicting financial distress in this context. However, Model 1 shows a relatively balanced classification performance and includes many features with significant importance values, making it advantageous for deeper analysis.

The Random Forest and Decision Tree models provide the best overall performance for financial distress prediction, achieving high accuracy and maintaining a strong balance between precision and recall. Overall, Random Forest consistently outperforms the other models in both datasets, showing high robustness and excellent performance. Decision Tree also performs well, especially with feature selection. SVM, however, is not recommended for this task without significant adjustments, as it exhibits poor performance with both the Altman and Zmijewski models. The feature selection appears to have positively impacted the models, improving their ability to detect distressed companies.

Chapter 5

Conclusion

Chapter 5 summarizes the entire research, identifies its limitations, and presents directions for future development.

5.1 Summary

Combining traditional distress prediction models with machine learning algorithms can enhance the ability to predict financial distress and provide early warning signs of bankruptcy. Traditional models, such as Altman's Z-score, are based on established financial ratios and provide a solid foundation for identifying distress. However, machine learning algorithms can improve these predictions by analyzing larger datasets, identifying complex patterns, and adapting to new, unseen data. This approach enables more precise and timely predictions, supported by dashboard visualizations, helping managers and investors take early action to reduce risk and prevent potential bankruptcy.

5.2 Future plan

In the future, to achieve more comprehensive and accurate findings, the study should consider incorporating alternative financial distress prediction models, such as the Ohlson O-score (1980), Springate S-Score (1978), and Blum's D-score (2004) [62] , along with model-free approaches for technique comparison [63]. It should also integrate a wider range of traditional, hybrid, and deep learning algorithms to improve prediction performance [64, 61]. Furthermore, the analysis should account for macroeconomic and market variables, as well as explore differences between pre- and post-COVID periods that may influence financial distress [59]. Examining firm performance across various business sectors can provide deeper insights. A sector-specific approach, in particular, will help managers better understand the critical roles of investment, financing, and operational strategies in driving business success.

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Appendix A

Thesis Timeline

Phases	No.	Sections	Details	September	October	November	December	January	February	March	April	May
1	1	Problem Definition	Define problem	x								
			Prepare knowledge	x								
	2	Research and Development	Research related work		x							
			Find suitable dataset		x							
			Propose baseline solution			x						
			Evaluation			x						
			Develop proposal				x					
2	3	Improve Solution	Research for better solutions					x				
			Prepare knowledge					x				
			Propose final solution						x			
			Final development and evaluation						x			
	4	Analysis and Visualization	Analyze dataset and result							x		
			Visualize result							x		
	5	Vertification and Report	Verify the work								x	
			Write graduation thesis								x	x

Appendix B

Performance Ratio

Index group	Attribute name	Code	Measurement
	EBIT Margin	X1	(Profit Before Tax + Interest) / Revenue
	EBITDA Margin	X2	(Profit Before Tax + Interest + Depreciation + Amortization) / Revenue
Solvency	Current ratio	X3	Current assets / Short-term liabilities
	Quick ratio	X4	(Current assets - inventories) / Current liabilities
	Cash ratio	X5	(Cash + Cash Equivalents) / Current Liabilities
	Working Capital-to-Total Assets (WCTA)	X6	(Current Assets - Current Liabilities) / Total Assets
	Debt-to-equity (D/E ratio)	X7	Total liabilities / Shareholder Equity
Operational Efficiency	Receivables turnover	X8	Net revenue/Receivables
	Inventory turnover	X9	Cost of Goods Sold / Average Inventory
	Accounts payable turnover (AP turnover)	X10	Supplier Credit Purchases / Average Accounts Payable
	Asset turnover	X11	Net Sales / Average total assets
	Current assets turnover	X12	Net Sales / Current assets
Profitability	Return on Assets (ROA)	X13	Net Income / Total Assets
	Return on Equity (ROE)	X14	Net Income / Shareholders' Equity
	Return on Invested Capital (ROIC)	X15	Net operating profit after tax (NOPAT) / Invested capital
	Gross Profit Margin	X16	Gross profit / Net Revenue
	Operating Profit Margin	X17	Operating Profit / Net Revenue
	Net Profit Margin	X18	Net profit / Net Revenue
Growth Capability	Capital Expenditure Growth (CAPEX) YoY	X19	ΔPP&E + Current Depreciation
	Revenue Growth 3Y	X20	
	Profit Before Tax (PBT) Growth	X21	
	Net Profit Growth	X22	
Risk Level	Total Debt on Total Assets	X23	Total Debt / Total Assets
Others	Basic Earnings Per Share (EPS)	X24	(Net Income - Preferred Dividends) / End-of-Period Common Shares Outstanding
	Basic Price-to-Earnings (P/E)	X25	Market value per share / Earnings per share
	Price-to-Book (P/B)	X26	Market price per share / Book value per share
	Enterprise Multiple (EV/EBITDA)	X27	Enterprise value / EBITDA
	Dividend yield	X28	Annual Dividends Per Share / Price Per Share